CIRC | The current teaching guide

Playbook A 2026-27 + Playbook B 2027-28

Open the route for your class. Find its lesson number below. Each complete lesson has 35-minute steps, teacher prompts, materials, evidence, adaptations and cleanup. Print only the lesson you need.

A has 31 meetings plus a buffer, including completed NeuroArt. B has 35 meetings plus a buffer. Shared lessons appear once. Grade 5 does KidWind and innovation; Grade 6 does CIRC Tank. This is not 74 extra meetings.

The lesson text comes directly from the same CIRC HQ source as the on-screen student and teacher views. The separate Classroom Toolkit covers Circuit Rescue, the logo-to-jingle companion and the optional NeuroArt finish check.

Dates and actual class progress stay with the teacher. No student names, class lists or private schedule are included.

Playbook A | 2026-27

31 meetings plus one buffer. From September 28, 2026 cycle days 1 to 3 have 31 visits left and days 4 and 5 have 30, so the buffer is only for days 1 to 3. Dates can shift. Mark a class by its record, never by the date.

Visit	Unit / season	Lesson number and path
1	Graphics and identity / Fall	L01 Grades 5 and 6: Neuro art and book checkout
2	Graphics and identity / Fall	L02 Grades 5 and 6: Canva logo 1: Start from a template
3	Graphics and identity / Fall	L03 Grades 5 and 6: Logo to jingle: Give your brand a sound
4	Build basics / Fall	L04 Grades 5 and 6: Cardboard joins and careful cutting
5	Build basics / Fall	L05 Grades 5 and 6: Make a package that protects
6	Build basics / Fall	L06 Grades 5 and 6: Paper Bridge
7	Two paths begin / Late fall	L07 Grade 5: Crash Lander L08 Grade 6: Tank 1: Notice a useful problem
8	Two paths begin / Late fall	L09 Grade 5: Cardboard Micro-Furniture L10 Grade 6: Tank 2: Listen before choosing
9	Flex A / Before Thanksgiving break	L11 Catch up: Catch up and finish L12 On track: Plan a small casting without pouring
10	KidWind and CIRC Tank / About December	L13 Grade 5: KidWind 1: Make the wind do work L14 Grade 6: Tank 3: Try three ideas
11	KidWind and CIRC Tank / About December	L15 Grade 5: KidWind 2: Change one thing L16 Grade 6: Tank 4: Plan the first build
12	KidWind and CIRC Tank / About December	L17 Grade 5: KidWind 3: Build the best set L18 Grade 6: Tank 5: Practice the tricky part
13	KidWind and CIRC Tank / About December	L19 Grade 5: KidWind 4: Tell what the wind taught us L20 Grade 6: Tank 6: Build version one
14	Grade 5 innovation, grade 6 Tank / Winter	L21 Grade 5: Human Program L22 Grade 6: Tank 7: Plan a fair test
15	Grade 5 innovation, grade 6 Tank / Winter	L23 Grade 5: Cargo Sorter L24 Grade 6: Tank 8: Test and record
16	Grade 5 innovation, grade 6 Tank / Winter	L25 Grade 5: Helping Hand L26 Grade 6: Tank 9: Improve one feature
17	Grade 5 innovation, grade 6 Tank / Winter	L27 Grade 5: Upcycle Lab L28 Grade 6: Tank 10: Let someone try
18	Grade 5 innovation, grade 6 Tank / Winter	L29 Grade 5: Accessibility Design Sprint L30 Grade 6: Tank 11: Use materials wisely
19	Grade 5 innovation, grade 6 Tank / Winter	L31 Grade 5: Moving Picture Machine L32 Grade 6: Tank 12: Name and package

Visit	Unit / season	Lesson number and path
20	Studio, pitch and share / Late winter	L33 Grade 5: Fix, Remix, or Invent Studio L34 Grade 6: Tank 13: Explain your idea
21	Studio, pitch and share / Late winter	L35 Grade 5: Innovation Studio 2: Change one thing and retest L36 Grade 6: Tank 14: Rehearse and refine
22	Studio, pitch and share / Late winter	L37 Grade 5: Innovation Share L38 Grade 6: Tank 15: Share and reflect
23	Water flow / Early spring	L39 Grades 5 and 6: Hydroponics Flow Lab
24	Tinkercad Skate Park / Spring	L40 Grades 5 and 6: Tinkercad 1: Move, turn, size
25	Tinkercad Skate Park / Spring	L41 Grades 5 and 6: Tinkercad 2: Holes and group
26	Tinkercad Skate Park / Spring	L42 Grades 5 and 6: Tinkercad 3: Align and mirror
27	Tinkercad Skate Park / Spring	L43 Grades 5 and 6: Tinkercad 4: Design a skate park piece
28	Tinkercad Skate Park / Spring	L44 Grades 5 and 6: Tinkercad 5: Fix, export, queue
29	Tinkercad Skate Park / Spring	L45 Grades 5 and 6: Tinkercad 6: Build day and pitch
30	Flex B and close / End of year	L11 Catch up: Catch up and finish L46 On track: Outdoor Microclimate Map
31	Flex B and close / End of year	L47 Grades 5 and 6: Reflect, reset, take home
32	Flex B and close / End of year	L11 Catch up: Catch up and finish

Playbook B | 2027-28

35 meetings plus one buffer, planned from this year's calendar of 35 or 36 visits per class. Check the 2027-28 calendar before relying on it. If a class has fewer visits, drop the flex on-track choices first.

Visit	Unit / season	Lesson number and path
1	Welcome and routines / Fall	L48 Grades 5 and 6: Welcome to CIRC
2	Welcome and routines / Fall	L49 Grades 5 and 6: Set Up Your Laptop
3	Welcome and routines / Fall	L50 Grades 5 and 6: Library launch: find it and ask a question
4	Research to design / Fall	L51 Grades 5 and 6: Fact card 1: Find and check a source
5	Research to design / Fall	L52 Grades 5 and 6: Fact card 2: Design it and share
6	Materials and water / Fall	L53 Grades 5 and 6: Mystery Materials
7	Materials and water / Fall	L54 Grades 5 and 6: Hull Design
8	Two paths begin / Late fall	L55 Grade 5: Seed Travelers L08 Grade 6: Tank 1: Notice a useful problem
9	Two paths begin / Late fall	L56 Grade 5: Paper Circuit Signal L10 Grade 6: Tank 2: Listen before choosing
10	Flex A / Before Thanksgiving break	L11 Catch up: Catch up and finish L57 On track: Earthquake Platform
11	KidWind and CIRC Tank / Late fall. Teacher places the four visits around December after checking next year's calendar	L13 Grade 5: KidWind 1: Make the wind do work L14 Grade 6: Tank 3: Try three ideas
12	KidWind and CIRC Tank / Late fall. Teacher places the four visits around December after checking next year's calendar	L15 Grade 5: KidWind 2: Change one thing L16 Grade 6: Tank 4: Plan the first build
13	KidWind and CIRC Tank / Late fall. Teacher places the four visits around December after checking next year's calendar	L17 Grade 5: KidWind 3: Build the best set L18 Grade 6: Tank 5: Practice the tricky part
14	KidWind and CIRC Tank / Late fall. Teacher places the four visits around December after checking next year's calendar	L19 Grade 5: KidWind 4: Tell what the wind taught us L20 Grade 6: Tank 6: Build version one
15	Grade 5 code and systems, grade 6 Tank / Winter	L58 Grade 5: Maze Coders L22 Grade 6: Tank 7: Plan a fair test
16	Grade 5 code and systems, grade 6 Tank / Winter	L59 Grade 5: Binary Beacons L24 Grade 6: Tank 8: Test and record
17	Grade 5 code and systems, grade 6 Tank / Winter	L60 Grade 5: Black Box Systems L26 Grade 6: Tank 9: Improve one feature

Visit	Unit / season	Lesson number and path
18	Grade 5 code and systems, grade 6 Tank / Winter	L61 Grade 5: Error-Proof Messages L28 Grade 6: Tank 10: Let someone try
19	Grade 5 code and systems, grade 6 Tank / Winter	L62 Grade 5: Biomimicry Grabber L30 Grade 6: Tank 11: Use materials wisely
20	Grade 5 code and systems, grade 6 Tank / Winter	L63 Grade 5: Rube Machine L32 Grade 6: Tank 12: Name and package
21	Studio, pitch and share / Late winter	L33 Grade 5: Fix, Remix, or Invent Studio L34 Grade 6: Tank 13: Explain your idea
22	Studio, pitch and share / Late winter	L35 Grade 5: Innovation Studio 2: Change one thing and retest L36 Grade 6: Tank 14: Rehearse and refine
23	Studio, pitch and share / Late winter	L37 Grade 5: Innovation Share L38 Grade 6: Tank 15: Share and reflect
24	Check the facts / Late winter	L64 Grades 5 and 6: Check the AI answer
25	Water and code / Spring	L65 Grades 5 and 6: Stormwater Rescue
26	Water and code / Spring	L66 Grades 5 and 6: Code 1: Draw with a sequence
27	Water and code / Spring	L67 Grades 5 and 6: Code 2: Loops make it move
28	Water and code / Spring	L68 Grades 5 and 6: Code 3: If this, then that
29	Water and code / Spring	L69 Grades 5 and 6: Micro:bit Sensor Station
30	Build and explore / Late spring	L70 Grades 5 and 6: Cardboard Arcade
31	Build and explore / Late spring	L71 Grades 5 and 6: Outdoor Habitat Helper
32	Build and explore / Late spring	L72 Grades 5 and 6: Outdoor Classroom Redesign
33	Flex B and close / End of year	L11 Catch up: Catch up and finish L73 On track: Mirror Maze
34	Flex B and close / End of year	L74 Grades 5 and 6: Demo Day
35	Flex B and close / End of year	L47 Grades 5 and 6: Reflect, reset, take home
36	Flex B and close / End of year	L11 Catch up: Catch up and finish

L01 | Neuro art and book checkout

GOAL: I can join curved lines, round their corners, and follow our book-checkout routine.

35 minutes. Lesson reference: year:circ-neuro

Before students arrive

Finished for this year. Reuse only for a class that missed it.

Materials

Watercolor paper; Black drawing markers; Washable color markers; Class artwork trays; Existing library books and checkout system

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | See our plan

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can join curved lines, round their corners, and follow our book-checkout routine.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-07 min | Watch the line

Students: Watch four curved lines cross. Turn one sharp corner into a smooth curve. Notice the large spaces left for color.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

07-20 min | Draw and round

Students: Write your name and class on the back. Draw four to six crossing curves. Round the sharp corners. Add dry color if time allows.

Teacher: Pause each group early. Ask what its next useful action will be. Use this short, dry-art version with checkout. Save wet blending for another contact using the existing neuro guide.

20-30 min | Choose your book

Students: Cap markers and place artwork in the class tray. When called, follow our library checkout routine. Return with your book and read quietly.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Point to one corner you rounded and show that your table is ready.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Artwork in the class tray, or the Paint file saved where the teacher showed.

Success check

Point to one corner you rounded and show that your table is ready.

Support and stretch

Grade 5: Four to six lines, large spaces, round every sharp corner.

Grade 6: Vary line widths and explain one color choice.

Another way to do it

Paper and markers with the same line rules.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L02 | Canva logo 1: Start from a template

GOAL: I can use a Canva logo template to make a logo and short slogan about something I am into.

35 minutes. Lesson reference: year:circ-canva-template

Before students arrive

Laptop cart charged and counted

Two or three real logos ready to show very small

Board checklist posted

Paper and pencils out for the fallback

Materials

Existing school devices on the cart; School Canva through resources.svsd.net; Two or three real logos to show small; Board checklist; Paper and pencils for the fallback

Safety

Use school Canva only. No new accounts, purchases or downloads. Share only with the teacher. Keep personal details private.

Run the period

00-04 min | Sign in

Students: Get your device. Open resources.svsd.net, then Canva. Sign in with your school account.

Teacher: Devices out. Help one table at a time with sign-in. If sign-in fails, switch that student to paper and pencil. Same steps.

04-08 min | Read tiny logos

Students: Look at the real logos, shown very small. Why can you still read them? Tell a partner one reason.

Teacher: Show two or three real logos very small. Listen for simple shapes, few colors and bold letters.

08-22 min | Make your logo

Students: Pick a logo template that fits one thing you are into. Change the colors, the icon and the font. Add a slogan of six words or fewer.

Teacher: Circulate. Ask which interest the logo shows and what they changed. No purchases. Skip paid elements and anything marked with a crown.

22-25 min | Small size check

Students: Zoom out until your logo is stamp size. Can you still read it? Fix one thing.

Teacher: Have students name the one thing they fixed.

25-29 min | Share with teacher

Students: Choose Share. Add your CIRC teacher's school account. It is turned in when your teacher can open it.

Teacher: It counts as turned in only when you can open it from your school account. Do not change public or link sharing. No downloads to the desktop.

29-32 min | Devices back

Students: Close Canva. Put your device back on the cart and plug it in. Line up at the door.

Teacher: Count the cart before the class leaves.

32-35 min | Exit question

Students: Answer at the line: what did the template do for you? Could you do that yourself next time?

Teacher: Accept a short spoken answer. Paper sketches go in the class tray for next visit.

What to collect or hear

Share with your CIRC teacher's school account. It is turned in when your teacher can open it. Keep public and link sharing unchanged.

Success check

Answer at the line: what did the template do for you? Could you do that yourself next time?

Support and stretch

Grade 5: Pick one interest. Change the colors, icon and font.

Grade 6: Explain why the slogan and icon match. Fix the weakest part at stamp size.

Another way to do it

Paper and pencil sketch with the same steps. Move it to Canva next visit.

Optional next challenge

Make a second color version or test your logo at an even smaller size.

Cleanup

Choose Share.

Add your CIRC teacher's school account.

It is turned in when your teacher can open it.

Close Canva.

Put your device back on the cart and plug it in.

Line up at the door.

Teacher notes

Graphics visit 1 of 2. NeuroArt is finished. This visit and the next replace the older Paint and digitize plan.

Turn in: Share with your CIRC teacher's school account. It is turned in when your teacher can open it. Keep public and link sharing unchanged. Check access from your school account. A name in the Share box alone is not enough.

If Canva or sign-in fails: paper and pencil sketch, same steps. Move it to Canva next visit.

Interests only. Nobody has to share anything personal. No names of classmates.

L03 | Logo to jingle: Give your brand a sound

GOAL: I can turn my first logo into a short slogan and a 10-15 second musical message.

35 minutes. Lesson reference: year:circ-logo-jingle

Before students arrive

Laptop cart charged and counted

First logos ready; NeuroArt and Logo 1 are finished

Open a teacher-operated jingle generator and test playback; prepare approved Canva audio as backup

Model a six-word slogan becoming a short jingle

Materials

Your saved first logo; School Canva through resources.svsd.net; Teacher-operated jingle generator or approved Canva audio; Headphones at a low volume; Paper and pencils for planning

Safety

Use your school Canva account. Your teacher operates external AI generators. Keep personal details out of prompts. No student AI accounts, purchases or public posting. Keep headphones low.

Run the period

00-04 min | Open your first logo

Students: Open resources.svsd.net, then Canva. Find your first logo. Today you are adding sound. Name the feeling or idea your logo should communicate.

Teacher: Show an existing logo and a short jingle. Today adds a sound identity to the first logo.

04-09 min | Write the slogan

Students: Write or choose three slogan ideas using six words or fewer. Try an action plus a benefit: Build bright. Think big. Choose one that matches your logo. Improve one word.

Teacher: Use the script helper from your teacher account if needed. Students select and revise the wording.

09-16 min | Plan your sound

Students: Choose a mood: upbeat, calm, funny or dramatic. Write your brand name, exact slogan and sound choice. Share that plan with your teacher. You do not need an AI account.

Teacher: Use the model prompt in teacher context. Generate selected examples while students prepare their Canva pages.

16-24 min | Generate and combine

Students: Watch the teacher turn a slogan into a jingle. Copy your first logo onto a Canva video page. Add the slogan. Use teacher-provided audio or an approved backing track. Aim for 10-15 seconds. Make the words easy to hear. Record the music source in your design.

Teacher: Use Suno for sung words or AI Jingle Maker for spoken words over music. If unavailable, use approved Canva audio and a spoken slogan. Students can finish their sound plan before audio arrives.

24-28 min | Listen and fix

Students: Play or perform your jingle for a partner. Can your partner repeat the slogan and explain how it matches the logo? Fix one unclear word, loud sound, or timing problem.

Teacher: Look for a matching logo, memorable slogan and clear sound choice. Check available audio or a live performance; retain unfinished generation plans.

28-31 min | Share with teacher

Students: Choose Share. Add your CIRC teacher school account. It is turned in when your teacher can open it. Save the logo, slogan, sound plan and any available audio.

Teacher: Do not change public or link sharing. Test a few shared designs from your school account.

31-35 min | Devices back

Students: Return headphones and lower the volume. Put your device back on the cart and plug it in. Clear your table.

Teacher: Count devices and collect paper versions. No extra performance is required after devices are put away.

What to collect or hear

Share with your CIRC teacher's school account. It is turned in when your teacher can open it. Keep public and link sharing unchanged. Include the first logo, slogan, sound plan and available audio. A live performance counts.

Success check

Play or perform your jingle for a partner. Can your partner repeat the slogan and explain how it matches the logo? Fix one unclear word, loud sound, or timing problem.

Support and stretch

Grade 5: One short slogan and a simple repeated beat.

Grade 6: Explain how the tempo and mood match your logo, and revise after listener feedback.

Another way to do it

Paper and pencil: show the first logo, write a slogan and perform it over a clapped beat. A partner may speak, or show the words.

Optional next challenge

Try a second tempo. Ask which version matches your logo, then keep the stronger one.

Cleanup

Choose Share.

Add your CIRC teacher school account. It is turned in when your teacher can open it.

Save the logo, slogan, sound plan and any available audio.

Return headphones and lower the volume.

Put your device back on the cart and plug it in.

Clear your table.

Teacher notes

This replaces Logo 2. NeuroArt/book checkout and Logo 1 are finished. Reuse the first logo.

Students plan the brand, slogan and sound. The teacher operates external AI tools and plays selected examples. Do not send under-13 students to generators or promise every group an individual render within this class.

Suno turns a slogan into a sung jingle. Use an existing teacher account if available. Free and paid plans have different download and use rights; check the active plan before exporting or publishing. Suno requires users to be at least 13, with parental consent under 18.

AI Jingle Maker has a script helper, AI voices and royalty-free music beds. Its instant result is a spoken slogan over music. The advertised starter pack is $39 plus tax for 100 voiceovers, not a free student tool. Custom sung jingles are a separate paid service. Use only if already available; no purchase is required for this lesson.

Model prompt: Create a 10-15 second original jingle for a fictional brand called [brand]. Use this slogan exactly: [slogan]. Mood: [mood]. Make the words easy to hear. No extra lyrics or imitation of an existing song. Generated duration may vary; trim in Canva.

If students need wording help, use the teacher-operated script helper to suggest three slogans under six words. Students choose and revise the one that best fits their first logo.

Before class, test the generator, playback and download access. Have approved Canva stock audio ready. Students can speak their slogan over that track if generation is unavailable. BeepBox is an optional original-music fallback.

Turn in the first logo, chosen slogan and sound plan in Canva. Add generated audio when available, or perform over the approved backing track. Share only with the teacher school account. Keep public and link sharing unchanged.

Royalty-free means use follows the applicable license; it does not mean every generator is free. Check the source and permitted use before publishing. In Canva, use teacher-approved stock audio rather than Popular Music.

L04 | Cardboard joins and careful cutting

GOAL: I can make a stable cardboard connection and test it before adding decoration.

35 minutes. Lesson reference: year:circ-cardboard

Before students arrive

Six table trays stocked, 30 scissors counted

Teacher cut station set. The knife stays with the teacher.

Materials

Clean cardboard; Paper; Pencils; Rulers; Student scissors; Masking tape; Reusable trays; Teacher-prepared pieces if needed; Optional approved ChompSaw station

Safety

Scissors are the default. A ChompSaw station opens only after school approval and the exact machine demonstration.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can make a stable cardboard connection and test it before adding decoration.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch the teacher cut away from the holding hand. Compare a tab, slot, and taped join. Watch the teacher test each join gently.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Sketch a small shape that stands by itself. Choose a tab, slot, or taped join. Cut with approved scissors or use teacher-prepared pieces. Assemble your shape.

Teacher: Pause each group early. Ask what its next useful action will be. ChompSaw purchase and operating approval are not assumed. Follow its manual and supervision rules if supplied. Teacher handles knife cuts and hot glue.

25-30 min | Check one thing

Students: Place your shape on the table and test its balance. Change one weak join. Compare the first and improved versions.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show one stable connection and explain why it holds.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Three labeled joint samples and one stable shape in the class tray.

Success check

Show one stable connection and explain why it holds.

Support and stretch

Grade 5: Build one stable shape using a demonstrated join.

Grade 6: Compare two join types and explain the tradeoff before choosing.

Another way to do it

Paper and cardstock tabs and slots work for every step.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L05 | Make a package that protects

GOAL: I can protect a model product, test the package, and improve one weak point.

35 minutes. Lesson reference: year:circ-packaging

Before students arrive

Six table trays stocked, 30 scissors counted

Model products and reclaimed boxes at each table

Handling test card posted

Materials

Reclaimed small boxes; Clean cardboard; Paper; String; Masking tape; Model products; Rulers; Scissors; Shared scale if available

Safety

Only use the teacher's handling test. Do not throw packages or test near people or equipment.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can protect a model product, test the package, and improve one weak point.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch a model product move inside a loose package. Notice padding, support, and an opening. Watch one teacher-approved handling test.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Choose the reusable model product provided. Sketch a package that opens without damage. Build using a material limit shown by your teacher.

Teacher: Pause each group early. Ask what its next useful action will be. Reuse catalog lesson 24, Package Rescue. A vacuum former is not needed for this complete lesson.

25-30 min | Check one thing

Students: Run the same gentle handling test twice. Check movement, damage, and ease of opening. Improve one weak point without adding unnecessary material.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Demonstrate how your package protects and opens, and explain one improvement.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Your package and one before and after change, shown to the teacher.

Success check

Demonstrate how your package protects and opens, and explain one improvement.

Support and stretch

Grade 5: Protect one object. Show one before and after change.

Grade 6: Compare protection, material use and how easy it is to open.

Another way to do it

Cardstock package tested by hand with the same handling card.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L06 | Paper Bridge

GOAL: Transform paper into a bridge that spans a gap and carries increasing loads.

35 minutes. Lesson reference: catalog:5

Before students arrive

Two support blocks per team at a fixed gap

Uniform load pieces counted

Record sheets out

Materials

Copy paper; Index cards; Tape; Two support blocks; Uniform load pieces; Recording sheet

Safety

Use only the provided loads and keep hands and feet clear beneath the test span.

Run the period

00-03 min | Get ready

Students: Place the support blocks on the marks. Count the load pieces.

Teacher: The material stays the same, but its shape can completely change its strength. A fair test changes the load while keeping the span and setup the same.

03-06 min | Test safely

Students: Keep hands below the table edge. Place only approved loads on the bridge.

Teacher: Compare a flat strip, folded beam, and rolled beam across the same gap.

06-11 min | Plan the span

Students: Choose a beam shape. Predict where the paper may bend. Sketch the bridge.

Teacher: Model adding load one unit at a time at the center.

11-22 min | Build the bridge

Students: Fold the paper into your chosen shape. Tape only the planned joints. Place the bridge across the fixed span.

Teacher: Stop tests before falling weights create a hazard.

22-29 min | Load the bridge

Students: Add one load piece. Record the supported total. Repeat until the bridge fails. Change one weak part.

29-32 min | Reset the test

Students: Return every load piece. Recycle damaged paper.

32-35 min | Exit check

Students: Write the highest supported load. Name your useful change.

What to collect or hear

Your record sheet: most load held, one rebuild, same test repeated.

Success check

Report the maximum load and explain how one shape helped carry it.

Support and stretch

Grade 5: Record the most load held with tally marks. Rebuild one part.

Grade 6: Record loads in a table. Explain which shape change mattered and why.

Another way to do it

The teacher runs one class test while teams predict and record.

Optional next challenge

Efficiency Engineer: Remove one piece of material while keeping the bridge able to hold its target load.

Cleanup

Stack reusable supports and load pieces, then place paper and tape scraps in the correct bins.

Teacher notes

The material stays the same, but its shape can completely change its strength.

A fair test changes the load while keeping the span and setup the same.

Compare a flat strip, folded beam, and rolled beam across the same gap.

Model adding load one unit at a time at the center.

Stop tests before falling weights create a hazard.

L07 | Crash Lander

GOAL: Design a landing system that reduces impact on a fragile payload during a controlled drop.

35 minutes. Lesson reference: catalog:8

Before students arrive

Six table trays stocked, 30 scissors counted

Drop zone marked. The teacher does every raised drop.

Reusable payloads counted

Materials

Small payload containers; Paper; Cardboard; Straws; Tape; Cotton; Reusable test payloads; Drop-zone team cards

Safety

Only the teacher releases from raised heights. Everyone stays behind the drop-zone boundary.

Run the period

00-03 min | Get ready

Students: Count your allowed materials. Place the payload beside the container. Pick up one drop-zone team card.

Teacher: The fall may be unavoidable, but the way force reaches the payload can be redesigned. A fair crash test uses the same payload, height, and release method.

03-06 min | Drop zone safety

Students: Stay behind the drop-zone line. Let the teacher release the lander. Do the waiting job on your team card. Move only when the teacher calls your team.

Teacher: Model crumple zones, padding, and load distribution with a sample capsule.

06-11 min | Plan the lander

Students: Mark where the payload needs support. Sketch one part that can bend safely.

Teacher: Set one controlled drop zone and release height.

11-22 min | Build the capsule

Students: Place the payload inside the container. Add only allowed materials. Check that no part can fall off.

Teacher: Inspect each design before approving a test.

22-29 min | Test and revise

Students: Predict the first failure point. Watch the controlled drop. Inspect the payload. Change one feature.

29-32 min | Reset the drop

Students: Return the reusable payload. Collect every loose part.

32-35 min | Exit check

Students: Draw the impact evidence. Label your useful change.

What to collect or hear

Impact evidence and one revision.

Success check

Point to the impact evidence and explain how the revision changed the force path.

Support and stretch

Support: Protect the payload with one idea, then improve it once.

Stretch: Explain how your change moved the force away from the payload.

Another way to do it

The teacher does low drops from table height, or design on paper.

Optional next challenge

Mass Limit: Reduce the capsule's material while keeping the payload protected in the same drop test.

Cleanup

Recover all capsule pieces and payloads before the next drop team enters the zone.

Teacher notes

The fall may be unavoidable, but the way force reaches the payload can be redesigned.

A fair crash test uses the same payload, height, and release method.

Model crumple zones, padding, and load distribution with a sample capsule.

Set one controlled drop zone and release height.

Inspect each design before approving a test.

L08 | Tank 1: Notice a useful problem

GOAL: I can notice a real, manageable problem and explain who it affects.

35 minutes. Lesson reference: year:tank-notice

Before students arrive

Teacher-approved problem pictures out

Class crate out, labeled with the class code only

Materials

Paper; Pencils; Teacher-approved problem pictures

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can notice a real, manageable problem and explain who it affects.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Compare a broad wish with one specific everyday problem. Notice who is affected and when it happens.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Look at the approved classroom examples. List or sketch three small problems. Choose one that can be explored with classroom materials.

Teacher: Pause each group early. Ask what its next useful action will be. Do not identify classmates as problems. Model a practical task or object that could work better.

25-30 min | Check one thing

Students: Tell a partner who has the problem. Explain when it happens. Narrow any idea that needs money, machinery, or changes outside our control.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Complete: Someone needs help with this problem when this happens.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Your problem sentence in the team folder: someone needs help with ___ when ___.

Success check

Complete: Someone needs help with this problem when this happens.

Support and stretch

Support: Choose from three problem picture cards.

Stretch: Name who it affects and how you know it is real.

Another way to do it

Talk it through while the teacher or a partner writes the sentence.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Find and describe a specific user problem with one direct observation.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L09 | Cardboard Micro-Furniture

GOAL: Build a miniature cardboard model of seating, storage, a work surface, or a quiet-space feature for a specific classroom user.

35 minutes. Lesson reference: catalog:12

Before students arrive

Six table trays stocked, 30 scissors counted

Paper user figures and user cards

Materials

Cardboard; Paper user figures; User cards; Rulers; Scissors; Tape; Paper fasteners; Grid paper; Markers

Safety

Cut only on the table, use fasteners as demonstrated, and keep models small enough to test safely at the workspace.

Run the period

00-03 min | Get ready

Students: Choose one user card. Choose one classroom location.

Teacher: Furniture is not just a shape. It must fit a person, a task, and a place. A small model lets designers test arrangement, stability, and access before using full-size materials. Present user cards for seating, storage, work-surface, and quiet-space needs.

03-06 min | Cut safely

Students: Cut only on the table. Keep fingers outside the cutting path. Carry scissors closed.

06-10 min | Plan for the user

Students: Draw the furniture feature. Label how the user reaches it. Mark the model size.

Teacher: Model measuring the intended location and labeling a miniature top, side, or opening dimension.

10-19 min | Build the model

Students: Cut the planned cardboard shapes. Use two connection methods. Keep the model table sized.

Teacher: Demonstrate a stability and access check using a paper user figure.

19-25 min | Test the user path

Students: Move the paper user toward the model. Check the reach point. Check the model stability.

25-29 min | Remove a problem

Students: Change one blocked or weak feature. Repeat the user path.

29-32 min | Sort the materials

Students: Close the scissors. Return usable cardboard. Store the model flat.

32-35 min | Exit check

Students: Point to the user feature. Explain the problem you changed.

What to collect or hear

A mini model tested with a paper user figure.

Success check

Name the user, location, and barrier, then demonstrate how the revised model improves access or function.

Support and stretch

Support: Pick a user card and build one feature.

Stretch: Test with two different users and adjust.

Another way to do it

Draw and label the design for the user.

Optional next challenge

Second User: Add one small change that makes the feature easier for a second user with a different reach or movement need.

Cleanup

Label models selected for review, flatten reusable cardboard, recycle clean scraps, and return tools by count.

Teacher notes

Furniture is not just a shape. It must fit a person, a task, and a place.

A small model lets designers test arrangement, stability, and access before using full-size materials.

Present user cards for seating, storage, work-surface, and quiet-space needs.

Model measuring the intended location and labeling a miniature top, side, or opening dimension.

Demonstrate a stability and access check using a paper user figure.

L10 | Tank 2: Listen before choosing

GOAL: I can ask a useful question and use the answer to improve my problem statement.

35 minutes. Lesson reference: year:tank-listen

Before students arrive

Reusable user cards at each table

Question starters on the board

Materials

Paper; Pencils; Reusable user cards

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can ask a useful question and use the answer to improve my problem statement.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Compare a leading question with an open question. Watch a short respectful interview.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Prepare two questions about the task or object. Ask an approved partner, or use the teacher's user card. Record the useful answer with a short note or drawing.

Teacher: Pause each group early. Ask what its next useful action will be. Use in-class role-play or approved users. Do not contact outside people or record private information.

25-30 min | Check one thing

Students: Compare the answer with your first idea. Change the problem statement if needed. Ask whether you are describing a need or jumping to a solution.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Share one thing you learned from the user and what it changed.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

A problem statement changed by one user's answer, in the team folder.

Success check

Share one thing you learned from the user and what it changed.

Support and stretch

Support: Use two question starters from the board.

Stretch: Ask a follow-up question and quote the answer.

Another way to do it

Role-play the user with a card.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Ask an open follow-up and separate what was observed from what was assumed.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L11 | Catch up and finish

GOAL: I can find my missing work, finish it, and show it is done.

35 minutes. Lesson reference: year:catch-up

Before students arrive

Class record of missed meetings

Catch-up jobs listed on the board

Materials from the missed lessons

Materials

Class record of missed meetings; Materials from the missed lessons; Team folders and class trays; Board list of catch-up jobs

Safety

Use only the materials for your catch-up job. The teacher runs any cutting or glue station.

Run the period

00-03 min | Check your list

Students: Look at the board. Find your missing piece of work.

Teacher: Post the catch-up jobs on the board before class.

03-04 min | Tell your partner

Students: Tell a partner what you will finish today.

Teacher: Listen for a clear, small job.

04-08 min | Pick your job

Students: Finish, redo a test, or help a teammate. Tell the teacher your pick.

Teacher: Write picks on a sticky note per table.

08-24 min | Catch up

Students: Work on your one job. Ask for help early.

Teacher: Visit students with the most missing first.

24-28 min | Check it

Students: Show your finished piece to a partner or the teacher.

Teacher: Mark the class record, not a public list.

28-33 min | Reset together

Students: Return tools and materials. Put finished work in the right tray. Clear your table.

Teacher: Check trays for the next class.

33-35 min | Show your result

Students: Say what you finished today.

Teacher: Accept a spoken answer.

What to collect or hear

The missing piece of work, shown to the teacher.

Success check

Say what you finished today.

Support and stretch

Grade 5: Finish one missing piece with a checklist.

Grade 6: Finish, then improve one earlier result.

Another way to do it

Help a teammate or improve one record sheet.

Optional next challenge

Help a teammate finish, or improve one label or record.

Cleanup

Return tools and materials.

Put finished work in the right tray.

Clear your table.

Teacher notes

Flex and buffer time. Use the class record to choose, never the date alone.

A class that missed a core meeting does that meeting first. Never start a multi-meeting unit in a flex slot.

Grade 5 and grade 6: same routine. Students pick one job from the board.

No-tech path: students who have nothing missing help a teammate or improve one record sheet.

L12 | Plan a small casting without pouring

GOAL: I can draw a useful small object, estimate its size, and compare possible materials.

35 minutes. Lesson reference: year:circ-resin-plan

Before students arrive

Paper, rulers and example molds or pictures

No resin out. This is dry planning only.

Materials

Paper; Pencils; Rulers; Calculators if available; Empty dry mold or dimensioned drawing; Teacher-cleared cured example or picture

Safety

This lesson stays dry. Students do not mix, pour, heat, demold, or handle uncured resin.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can draw a useful small object, estimate its size, and compare possible materials.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Look at a dry mold or its measured drawing. Identify the inside space that shapes the object. Compare a cardboard model with a teacher-cleared cured example or picture.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Sketch a small token, tag, or game piece. Measure a rectangular practice cavity or its drawing. Use length, width, and depth to estimate its volume.

Teacher: Pause each group early. Ask what its next useful action will be. Only a small teacher trial is planned, pending the actual product, school procedure, and Home Depot. No student casting rollout is approved.

25-30 min | Check one thing

Students: Compare making your design from cardboard and casting material. Name one reason for each choice. Improve the design using the teacher's size limit.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show a labeled design and explain why your chosen material fits its use.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Your labeled drawing and size estimate.

Success check

Show a labeled design and explain why your chosen material fits its use.

Support and stretch

Grade 5: Teacher-supported measuring and a labeled drawing.

Grade 6: Calculate the volume and explain an irregular mold.

Another way to do it

The whole lesson is paper based.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L13 | KidWind 1: Make the wind do work

GOAL: I can build blades that turn wind into work and measure how far they lift a load.

35 minutes. Lesson reference: year:kidwind-lift

Before students arrive

Preparation required before the unit: build and pilot one working station from the parts actually on hand. Check free turning, reliable winding and a clear lift path.

Pilot the load: choose a cup load the prototype can raise, and write down the exact load.

Tape the start line, a mark 20 cm above it, the fan distance and the stand-back line.

Time one full team trial and swap. If it does not fit, use the shorter trial counts in the teacher notes.

Materials

One teacher-run test station, built and piloted first from parts on hand; One guarded fan at one fixed setting, with tape marks and a ruler; A smooth, blunt shaft turning in a larger straw or tube, with a cork or foam hub; String on a fixed spool, one small cup and the same load pieces every trial; Cardstock or index cards, reusable blade supports, tape and student scissors; A marked blade-angle example, timer, shallow catch tray, team folders and record sheets; Printed practice data table for the no-fan paper path

Safety

Only the teacher runs the fan and prepares any holes with pointed tools. Stay out of the rotor's spin path. Fan off before anyone touches blades, the cup or the string. Blades are paper or cardstock only. Shaft ends are blunt.

Prepare and test this first

Build and pilot one working station, choose and record the load, and time one trial before teaching. Nothing here confirms that preparation is done.

Run the period

00-02 min | Get ready

Students: Sit with your team. Read the goal: I can make the wind do work.

Teacher: Say the goal. Show the station lifting the cup once.

02-03 min | Tell your partner

Students: Tell your partner one place you feel strong wind.

Teacher: Take one or two answers. Link wind to pushing.

03-08 min | Station safety

Students: Watch the cup rise from the start line. Stay out of the spin path. Only the teacher turns the fan on or off.

Teacher: Show the start line, the 20 cm mark and the stand-back line.

08-22 min | Build a rotor set

Students: Cut three blades from cardstock. Tape each blade to a blade support. Set each blade at the same angle as the example.

Teacher: Check that every blade is secure before it goes to the station.

22-30 min | Test at the station

Students: Your team gets one 10 second trial. Watch the cup rise. Record the centimeters lifted.

Teacher: Call teams in order. Write each result on the class chart.

30-33 min | Reset the station

Students: Fan off first. Rotor sets go flat in your team folder. Cup and load pieces go back in the station tray.

Teacher: Unplug the fan. Check that every station part is back.

33-35 min | Show your result

Students: Say how many centimeters your blades lifted the cup. Name one thing you might change.

Teacher: Accept a spoken answer or pointing at the chart.

What to collect or hear

Your team's centimeters lifted in 10 seconds, on your record sheet and the class chart.

Success check

Say how many centimeters your blades lifted the cup. Name one thing you might change.

Support and stretch

Support: Use the blade-angle example exactly.

Stretch: Predict the centimeters before you test.

Another way to do it

No-fan paper path: follow the paper steps and the labeled practice data below. It uses the same measurement but does not claim you built or tested a windmill.

Optional next challenge

Cut a spare set of blades in a new shape and label what you would test next.

Cleanup

Fan off first.

Rotor sets go flat in your team folder.

Cup and load pieces go back in the station tray.

Teacher notes

Grade 5 only, first of four KidWind visits. Grade 6 runs CIRC Tank in the same weeks.

Before the unit, build and pilot one working station. Teams make rotor sets and share that one station.

Test rule: cup starts at the start line with the recorded load. Fan on for 10 seconds. Record centimeters lifted, 0 to 20. Stop early at the 20 cm mark so the cup never winds into the shaft. Fan off before any change or reset. Fan setting, distance, cup, load, shaft and spool never change.

Adapted from KidWind's MacGyver Windmills activity. The four 35-minute visits are a CIRC adaptation, not an official KidWind lesson. No purchase, owned equipment or competition is implied.

Complete paper path

Draw and label the blades, hub, shaft, string and cup.

Read the practice table.

Explain what one result means in centimeters.

Paper evidence

A labeled drawing and a correct reading of one result.

Practice data

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Practice design	Only stated change	Trial 1	Trial 2	Trial 3
A	Baseline	5	10	5
B	Blade angle only	10	15	10
C	Blade size only, compared with A	10	10	15

L14 | Tank 3: Try three ideas

GOAL: I can sketch different solutions and choose one using a clear reason.

35 minutes. Lesson reference: year:tank-options

Before students arrive

Criteria cards and rulers out

Materials

Paper; Pencils; Rulers; Reusable criteria cards

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can sketch different solutions and choose one using a clear reason.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: See three different ways to solve one small problem. Compare usefulness, simplicity, and available materials.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Sketch three different solutions. Label the part that does the work. Circle one idea you can test with classroom materials.

Teacher: Pause each group early. Ask what its next useful action will be.

25-30 min | Check one thing

Students: Show the ideas to a partner. Name one strength and one concern for the chosen idea. Revise the sketch before building.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show three ideas and explain why one is the best first test.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Three sketches, one chosen idea and the reason.

Success check

Show three ideas and explain why one is the best first test.

Support and stretch

Support: Sketch two ideas, then one more with a partner.

Stretch: Score each idea against two criteria before choosing.

Another way to do it

Describe three ideas aloud while a partner sketches.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Use two chosen criteria to compare all three ideas.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L15 | KidWind 2: Change one thing

GOAL: I can change one thing about my blades and keep everything else the same.

35 minutes. Lesson reference: year:kidwind-variable

Before students arrive

Preparation required before the unit: build and pilot one working station from the parts actually on hand. Check free turning, reliable winding and a clear lift path.

Pilot the load: choose a cup load the prototype can raise, and write down the exact load.

Tape the start line, a mark 20 cm above it, the fan distance and the stand-back line.

Time one full team trial and swap. If it does not fit, use the shorter trial counts in the teacher notes.

Materials

One teacher-run test station, built and piloted first from parts on hand; One guarded fan at one fixed setting, with tape marks and a ruler; A smooth, blunt shaft turning in a larger straw or tube, with a cork or foam hub; String on a fixed spool, one small cup and the same load pieces every trial; Cardstock or index cards, reusable blade supports, tape and student scissors; A marked blade-angle example, timer, shallow catch tray, team folders and record sheets; Printed practice data table for the no-fan paper path

Safety

Only the teacher runs the fan and prepares any holes with pointed tools. Stay out of the rotor's spin path. Fan off before anyone touches blades, the cup or the string. Blades are paper or cardstock only. Shaft ends are blunt.

Prepare and test this first

Build and pilot one working station, choose and record the load, and time one trial before teaching. Nothing here confirms that preparation is done.

Run the period

00-02 min | Get ready

Students: Find your team folder and your Set A rotor. Read the goal: change one thing only.

Teacher: Say the goal. Check that every team has Set A.

02-03 min | Read the chart

Students: Look at last time's class chart.

Teacher: Point to the highest and lowest results in centimeters.

03-07 min | Watch a fair test

Students: Watch the teacher change only the blade angle. Notice what stays the same.

Teacher: Name what stays the same: fan, distance, cup, load, shaft, spool.

07-19 min | Build Set B

Students: Pick one thing to change: number, size, angle or shape. Keep everything else the same as Set A. Write the one change on your record sheet.

Teacher: Ask each team to name its one change before it tests.

19-30 min | Test A and B

Students: Test Set A once. Test Set B once. Record centimeters lifted for each trial.

Teacher: Run both trials for a team back to back. Fan off for each swap.

30-33 min | Reset the station

Students: Fan off first. Rotor sets go flat in your team folder. Cup and load pieces go back in the station tray.

Teacher: Unplug the fan. Check that every station part is back.

33-35 min | Share

Students: Tell the class which set lifted the cup higher and by how many centimeters.

Teacher: Add each team's change to the class chart. Call them first results.

What to collect or hear

Set A and Set B results in centimeters, with the one change named. These are first results.

Success check

Tell the class which set lifted the cup higher and by how many centimeters.

Support and stretch

Support: Choose the change from two cards.

Stretch: Explain why everything else must stay the same.

Another way to do it

No-fan paper path: follow the paper steps and the labeled practice data below. It uses the same measurement but does not claim you built or tested a windmill.

Optional next challenge

Predict which change would help most next time and write why.

Cleanup

Fan off first.

Rotor sets go flat in your team folder.

Cup and load pieces go back in the station tray.

Teacher notes

Grade 5 only, second KidWind visit. Teams need their Set A rotor from last time.

One station: each team gets one Set A trial and one Set B trial. These are first results. Visit 3 repeats trials.

Test rule: cup starts at the start line with the recorded load. Fan on for 10 seconds. Record centimeters lifted, 0 to 20. Stop early at the 20 cm mark so the cup never winds into the shaft. Fan off before any change or reset. Fan setting, distance, cup, load, shaft and spool never change.

Adapted from KidWind's MacGyver Windmills activity. The four 35-minute visits are a CIRC adaptation, not an official KidWind lesson. No purchase, owned equipment or competition is implied.

Complete paper path

Compare design A with design B.

Name the one stated change.

List what the test must keep the same.

Read all three trials for each.

Paper evidence

A one-change sentence and a fair-test checklist.

Practice data

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Practice design	Only stated change	Trial 1	Trial 2	Trial 3
A	Baseline	5	10	5
B	Blade angle only	10	15	10
C	Blade size only, compared with A	10	10	15

L16 | Tank 4: Plan the first build

GOAL: I can make a labeled plan and choose a small, testable first version.

35 minutes. Lesson reference: year:tank-plan

Before students arrive

Six table trays stocked, 30 scissors counted

Class crate out, labeled with the class code only

Materials

Clean cardboard; Paper; Pencils; Rulers; Student scissors; Masking tape; Reusable trays

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can make a labeled plan and choose a small, testable first version.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch a sketch become a parts list. Separate essential working parts from decoration.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Draw the first version from a useful view. Label its working parts and approximate sizes. Choose materials and agree on the first build jobs.

Teacher: Pause each group early. Ask what its next useful action will be. Limit each team to a small tray of materials. No shopping or new equipment is needed for the first prototype.

25-30 min | Check one thing

Students: Explain how a user will try the model. Remove any part that does not help the first test. Show the teacher your plan before collecting supplies.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show a labeled plan, short materials list, and one thing your model must do.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

A labeled plan with a materials list.

Success check

Show a labeled plan, short materials list, and one thing your model must do.

Support and stretch

Support: Label three parts.

Stretch: Add measurements and stay inside the size limit.

Another way to do it

Plan on paper only.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Include a material limit, approximate dimensions, and a measurable success criterion.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L17 | KidWind 3: Build the best set

GOAL: I can use class evidence to build a better set of blades.

35 minutes. Lesson reference: year:kidwind-best

Before students arrive

Preparation required before the unit: build and pilot one working station from the parts actually on hand. Check free turning, reliable winding and a clear lift path.

Pilot the load: choose a cup load the prototype can raise, and write down the exact load.

Tape the start line, a mark 20 cm above it, the fan distance and the stand-back line.

Time one full team trial and swap. If it does not fit, use the shorter trial counts in the teacher notes.

Materials

One teacher-run test station, built and piloted first from parts on hand; One guarded fan at one fixed setting, with tape marks and a ruler; A smooth, blunt shaft turning in a larger straw or tube, with a cork or foam hub; String on a fixed spool, one small cup and the same load pieces every trial; Cardstock or index cards, reusable blade supports, tape and student scissors; A marked blade-angle example, timer, shallow catch tray, team folders and record sheets; Printed practice data table for the no-fan paper path

Safety

Only the teacher runs the fan and prepares any holes with pointed tools. Stay out of the rotor's spin path. Fan off before anyone touches blades, the cup or the string. Blades are paper or cardstock only. Shaft ends are blunt.

Prepare and test this first

Build and pilot one working station, choose and record the load, and time one trial before teaching. Nothing here confirms that preparation is done.

Run the period

00-02 min | Get ready

Students: Sit with your team. Read the goal: use evidence to build better.

Teacher: Say the goal. Put the class chart on the screen.

02-03 min | Find your records

Students: Find your Set A and Set B results.

Teacher: Hand out folders if needed.

03-08 min | Read the class chart

Students: Look for the sets that lifted the cup highest. What did those blades have in common?

Teacher: Circle the top three results. Ask what they share.

08-20 min | Build your best set

Students: Use the evidence to design one best set. Say which chart result supports your choice.

Teacher: Listen for evidence, not favorites.

20-30 min | Repeat fair trials

Students: Run two or three 10-second trials, as your teacher directs. Record every result in centimeters, even the low ones.

Teacher: Keep the fan, distance, cup and load the same for every team.

30-33 min | Reset the station

Students: Fan off first. Rotor sets go flat in your team folder. Cup and load pieces go back in the station tray.

Teacher: Unplug the fan. Check that every station part is back.

33-35 min | Compare

Students: Compare your best set with your first set, in centimeters.

Teacher: Accept a number and a reason.

What to collect or hear

Your best set's results from two or three 10-second trials, as your teacher directs, in centimeters and compared with your first set.

Success check

Compare your best set with your first set, in centimeters.

Support and stretch

Support: Copy one feature from a top result.

Stretch: Combine two features from different top results.

Another way to do it

No-fan paper path: follow the paper steps and the labeled practice data below. It uses the same measurement but does not claim you built or tested a windmill.

Optional next challenge

Test one blade angle change on your best set and record it separately.

Cleanup

Fan off first.

Rotor sets go flat in your team folder.

Cup and load pieces go back in the station tray.

Teacher notes

Grade 5 only, third KidWind visit. The class chart is the evidence for today's design.

Each team runs three trials in a row without changing the rotor. If the pilot shows this does not fit one station, run two and say so.

Test rule: cup starts at the start line with the recorded load. Fan on for 10 seconds. Record centimeters lifted, 0 to 20. Stop early at the 20 cm mark so the cup never winds into the shaft. Fan off before any change or reset. Fan setting, distance, cup, load, shaft and spool never change.

Adapted from KidWind's MacGyver Windmills activity. The four 35-minute visits are a CIRC adaptation, not an official KidWind lesson. No purchase, owned equipment or competition is implied.

Complete paper path

Compare design A with design C.

Choose a design to test next.

Use two table numbers to explain your choice.

Draw the next test.

Paper evidence

A prediction backed by the table. It is not a measured new result.

Practice data

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Practice design	Only stated change	Trial 1	Trial 2	Trial 3
A	Baseline	5	10	5
B	Blade angle only	10	15	10
C	Blade size only, compared with A	10	10	15

L18 | Tank 5: Practice the tricky part

GOAL: I can practice one connection or moving part before using it in my prototype.

35 minutes. Lesson reference: year:tank-joins

Before students arrive

Six table trays stocked, 30 scissors counted

Craft sticks, string, fasteners and bands in trays

Teacher cut and glue station set

Materials

Clean cardboard; Paper; Pencils; Rulers; Student scissors; Masking tape; Reusable trays; Craft sticks; String; Paper fasteners; Rubber bands

Safety

Keep moving parts at table level. No launching or pinching tests. Use only teacher-approved tools.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can practice one connection or moving part before using it in my prototype.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch a tab, hinge, slider, or string pull. Identify where friction, looseness, or weakness could occur.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Choose the hardest working part in your plan. Build a small practice sample using scrap. Try the movement or connection several times.

Teacher: Pause each group early. Ask what its next useful action will be.

25-30 min | Check one thing

Students: Name the first weak or unreliable point. Change one feature. Keep the stronger sample as a reference.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Demonstrate your practice part and name the improvement.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

An improved practice sample in the class crate.

Success check

Demonstrate your practice part and name the improvement.

Support and stretch

Support: Copy one demonstrated join.

Stretch: Test two joins and keep the stronger one.

Another way to do it

Tape and tab joins only.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Compare two mechanisms against the same job and select one using test evidence.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L19 | KidWind 4: Tell what the wind taught us

GOAL: I can explain what we changed, what happened, and the proof.

35 minutes. Lesson reference: year:kidwind-share

Before students arrive

Preparation required before the unit: build and pilot one working station from the parts actually on hand. Check free turning, reliable winding and a clear lift path.

Pilot the load: choose a cup load the prototype can raise, and write down the exact load.

Tape the start line, a mark 20 cm above it, the fan distance and the stand-back line.

Time one full team trial and swap. If it does not fit, use the shorter trial counts in the teacher notes.

Materials

One teacher-run test station, built and piloted first from parts on hand; One guarded fan at one fixed setting, with tape marks and a ruler; A smooth, blunt shaft turning in a larger straw or tube, with a cork or foam hub; String on a fixed spool, one small cup and the same load pieces every trial; Cardstock or index cards, reusable blade supports, tape and student scissors; A marked blade-angle example, timer, shallow catch tray, team folders and record sheets; Printed practice data table for the no-fan paper path; Three picture prompt cards: what we changed, what happened, our proof

Safety

Only the teacher runs the fan and prepares any holes with pointed tools. Stay out of the rotor's spin path. Fan off before anyone touches blades, the cup or the string. Blades are paper or cardstock only. Shaft ends are blunt.

Prepare and test this first

Build and pilot one working station, choose and record the load, and time one trial before teaching. Nothing here confirms that preparation is done.

Run the period

00-02 min | Get ready

Students: Quick recap: what did our blades do?

Teacher: Recap the chart in one minute for classes back from break.

02-03 min | Find your records

Students: Get your folder and your best set.

Teacher: Hand out the picture prompt cards.

03-11 min | Final test

Students: Run your best set for one 10 second trial. Record the centimeters lifted.

Teacher: Run the station in team order.

11-23 min | Plan your talk

Students: Plan a 30 to 45 second talk. Say what we changed, what happened, and our proof in centimeters.

Teacher: Check that every team has a centimeter result as proof.

23-30 min | Share your talk

Students: Give your talk to another team. Listen and ask one question.

Teacher: Pair teams. Time each talk.

30-33 min | Reset the station

Students: Fan off first. Rotor sets go flat in your team folder. Cup and load pieces go back in the station tray.

Teacher: Unplug the fan. Check that every station part is back.

33-35 min | Show your result

Students: Say one thing the wind taught you.

Teacher: Accept a spoken answer.

What to collect or hear

A 30 to 45 second talk with a centimeter result as proof.

Success check

Say one thing the wind taught you.

Support and stretch

Support: Use the three picture prompt cards.

Stretch: Add a comparison to your first test.

Another way to do it

No-fan paper path: follow the paper steps and the labeled practice data below. It uses the same measurement but does not claim you built or tested a windmill.

Optional next challenge

Draw a bar chart of your three trials from last time.

Cleanup

Fan off first.

Rotor sets go flat in your team folder.

Cup and load pieces go back in the station tray.

Teacher notes

Grade 5 only, last KidWind visit. It works after a winter break with a short recap.

Close with: in the innovation studio you will test your own idea the same way.

Test rule: cup starts at the start line with the recorded load. Fan on for 10 seconds. Record centimeters lifted, 0 to 20. Stop early at the 20 cm mark so the cup never winds into the shaft. Fan off before any change or reset. Fan setting, distance, cup, load, shaft and spool never change.

Adapted from KidWind's MacGyver Windmills activity. The four 35-minute visits are a CIRC adaptation, not an official KidWind lesson. No purchase, owned equipment or competition is implied.

Complete paper path

Plan a 30 to 45 second talk.

Say the change you propose and the practice evidence.

Say what a real test must check.

Call the numbers practice data.

Paper evidence

A talk or written card that calls the numbers practice data.

Practice data

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Practice design	Only stated change	Trial 1	Trial 2	Trial 3
A	Baseline	5	10	5
B	Blade angle only	10	15	10
C	Blade size only, compared with A	10	10	15

L20 | Tank 6: Build version one

GOAL: I can build a rough model that shows my idea's main job.

35 minutes. Lesson reference: year:tank-prototype

Before students arrive

Six table trays stocked, 30 scissors counted

Class crate out, labeled with the class code only

Cell gauge ready: 21 by 14 by 12 cm

Materials

Clean cardboard; Paper; Pencils; Rulers; Student scissors; Masking tape; Reusable trays; Craft sticks; String; Paper fasteners

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can build a rough model that shows my idea's main job.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch the main working part being built first. Notice that a rough prototype can answer a useful question.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Build the essential working part from your plan. Add only the support needed to try it. Keep a sketch or picture of version one using the approved classroom method.

Teacher: Pause each group early. Ask what its next useful action will be. A failed first attempt is useful evidence. Do not add decoration before the working part is testable.

25-30 min | Check one thing

Students: Ask a partner to identify the model's job. Try the intended action once. Mark one feature that needs a test next time.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show version one doing its main job, or identify the exact part that needs repair.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Version one in your crate cell with its folder.

Success check

Show version one doing its main job, or identify the exact part that needs repair.

Support and stretch

Support: Build only the part that does the main job.

Stretch: Build to the planned measurements.

Another way to do it

A paper model or drawing that shows the main job.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Build a working feature within the agreed size and material limits.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L21 | Human Program

GOAL: Create and debug an exact sequence of commands that guides a human robot through a task.

35 minutes. Lesson reference: catalog:14

Before students arrive

Command cards and grid mats

Safe classroom objects

Materials

Command cards; Grid mats; Safe classroom objects; Pencils; Clipboards; Short-task cards

Safety

Human robots move at walking speed and stop immediately at any obstacle or stop command.

Run the period

00-03 min | Get ready

Students: Choose one short task card. Place the grid mat on the floor.

Teacher: Computers do exactly what the instructions say, not what the programmer hoped they meant. A bug is useful evidence that a command needs to be more precise.

03-06 min | Robot safety

Students: Move only at walking speed. Stop at every obstacle. Stop when anyone says stop.

Teacher: Model ambiguous and precise commands with a simple desk task.

06-11 min | Write commands

Students: Write one command per line. Use only approved command words.

Teacher: Introduce the approved move, turn, pick up, and place command set.

11-18 min | Run the program

Students: Give the commands to the human robot. Watch each command run exactly. Mark the first wrong result.

Teacher: Pause any program that would create a collision.

18-24 min | Debug one line

Students: Rewrite the first wrong command. Run the complete program again.

24-29 min | Switch roles

Students: Choose a new human robot. Run the corrected program once.

29-32 min | Reset the grid

Students: Stack the command cards. Roll the grid mat.

32-35 min | Exit check

Students: Copy the corrected command. Name the result it changed.

What to collect or hear

The first bug and the command change that fixed it.

Success check

Show the original bug and the exact command revision that fixed it.

Support and stretch

Support: Use four command cards.

Stretch: Find the shortest program that works.

Another way to do it

This lesson is already device free.

Optional next challenge

Loop Finder: Replace a repeated command sequence with one clear repeat instruction.

Cleanup

Reset grid objects to the starting diagram and return command cards in order.

Teacher notes

Computers do exactly what the instructions say, not what the programmer hoped they meant.

A bug is useful evidence that a command needs to be more precise.

Model ambiguous and precise commands with a simple desk task.

Introduce the approved move, turn, pick up, and place command set.

Pause any program that would create a collision.

L22 | Tank 7: Plan a fair test

GOAL: I can plan a repeatable test that checks what my design claims.

35 minutes. Lesson reference: year:tank-test-plan

Before students arrive

Prototypes out of crates

Test objects and timers ready

Materials

Prototype; Paper; Pencils; Ruler or timer if needed; Reusable test objects

Safety

Only conduct teacher-approved tests at table level. No people, valuables, or live animals are test objects.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can plan a repeatable test that checks what my design claims.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Compare two tests that change several things. Watch a fair test keep the object, starting point, and action consistent.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Write or draw what the product claims to do. Choose one thing to count, measure, or observe. Plan three trials with the same starting conditions.

Teacher: Pause each group early. Ask what its next useful action will be.

25-30 min | Check one thing

Students: Walk through the test without rushing. Ask a partner what must stay the same. Get the teacher's approval before the real trials.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Explain your test, what stays the same, and what result would count as success.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

A written test plan: what changes and what stays the same.

Success check

Explain your test, what stays the same, and what result would count as success.

Support and stretch

Support: Fill in a test plan frame.

Stretch: Add a success number you can measure.

Another way to do it

Explain the test aloud to the teacher.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Choose a numerical measure and a success threshold before testing.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L23 | Cargo Sorter

GOAL: Design a mechanical sorting system that routes model cargo by one measurable property.

35 minutes. Lesson reference: catalog:19

Before students arrive

Six table trays stocked, 30 scissors counted

Mixed model cargo in batches

Materials

Mixed model cargo; Cardboard; Ramps; Cups; Craft sticks; Tape; Optional magnets

Safety

Keep magnets away from personal electronics and feed cargo without putting fingers inside a moving path.

Run the period

00-03 min | Get ready

Students: Place mixed cargo in the starting cup. Choose two ending bins.

Teacher: A sorter needs a clear input, a decision, and an output for every item. If cargo gets stuck, that is evidence about the geometry of the system.

03-06 min | Sorter safety

Students: Keep fingers outside every moving path. Keep magnets away from electronics. Feed one cargo item at a time.

Teacher: Model a ramp gate that separates objects by size.

06-10 min | Plan the sort

Students: Choose one sorting rule. Draw the travel path. Label both ending bins.

Teacher: Demonstrate feeding one item at a time from the same start point.

10-19 min | Build the sorter

Students: Build the starting ramp. Add one part that sends pieces to a bin. Place the bins under the exits.

Teacher: Set a target for both accuracy and jam-free operation.

19-26 min | Run the batch

Students: Feed one mixed item. Record its ending bin. Repeat for all test pieces.

26-30 min | Fix the errors

Students: Change the sorting part. Test all pieces again.

30-32 min | Reset the cargo

Students: Count every cargo item. Return magnets to the marked box.

32-35 min | Exit check

Students: Write the final error count. Name the sorting rule.

What to collect or hear

Batch accuracy and the fix that stopped the most common error.

Success check

Report batch accuracy and show the revision that prevented the most common error.

Support and stretch

Support: Sort by one property you can see.

Stretch: Report batch accuracy as a fraction.

Another way to do it

Sort by hand with a written rule first.

Optional next challenge

Third Destination: Add a third output without reducing the accuracy of the original two groups.

Cleanup

Recount each cargo type, return magnets separately, and flatten reusable ramp materials.

Teacher notes

A sorter needs a clear input, a decision, and an output for every item.

If cargo gets stuck, that is evidence about the geometry of the system.

Model a ramp gate that separates objects by size.

Demonstrate feeding one item at a time from the same start point.

Set a target for both accuracy and jam-free operation.

L24 | Tank 8: Test and record

GOAL: I can run the same test three times and record what actually happens.

35 minutes. Lesson reference: year:tank-testing

Before students arrive

Test objects and record sheets at each table

Materials

Prototype; Approved test objects; Paper; Pencils; Ruler or timer if needed

Safety

Use the approved test only. Stop if a part breaks sharply or the setup becomes unstable.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can run the same test three times and record what actually happens.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch one trial from the same starting position. See a record include an unsuccessful result honestly.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Reset the test to the agreed starting position. Run three trials and record each result. Keep the design unchanged until the trials are finished.

Teacher: Pause each group early. Ask what its next useful action will be.

25-30 min | Check one thing

Students: Compare all three results. Point to the most useful pattern or failure. Choose one problem to fix next.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show three results and identify one specific design problem.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Three recorded trials in the team folder.

Success check

Show three results and identify one specific design problem.

Support and stretch

Support: Record with tally marks.

Stretch: Find the middle result and name one surprise.

Another way to do it

The teacher runs the test while the team records.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Use measured results and explain variation between the trials.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L25 | Helping Hand

GOAL: Build a hand-operated tool that safely grips and moves an object beyond normal reach.

35 minutes. Lesson reference: catalog:23

Before students arrive

Six table trays stocked, 30 scissors counted

Lightweight test objects and role cards

Materials

Cardboard; Craft sticks; String; Rubber bands; Brass fasteners; Foam pads; Lightweight test objects; Scissors; Role cards; Build-lead marker; Test stand

Safety

Grip only approved lightweight objects and keep the device away from faces, bodies, and breakable items.

Run the period

00-03 min | Get ready

Students: Read one role card. Join the one shared educational prototype team.

Teacher: Assistive design begins with what a person wants to do, not with assumptions about the person. A useful gripper balances reach, control, force, and comfort. Assign the plan, parts, build, test, fix, and record role cards before work starts.

03-07 min | Build safely

Students: Cut only on the table. Keep fingers outside the cutting path. Carry scissors closed. Keep the prototype away from bodies.

Teacher: Model respectful need-first language and a simple linkage gripper.

07-12 min | Plan the grip

Students: Choose one lightweight test object. Sketch the handle. Sketch the moving connection. Mark the soft part that holds the object.

Teacher: Demonstrate testing with lightweight objects only.

12-20 min | Build one shared part

Students: Do only the work on your role card. Put your finished work in the team tray. Show it to the build lead.

Teacher: Check every moving joint and handle before use.

20-26 min | Test on the stand

Students: The test team places the build on the test stand. The test team moves one approved object. The record team writes the result with no student name.

26-30 min | Improve control

Students: The fix team changes one moving or soft part. The test team repeats the stand test.

30-32 min | Reset the build

Students: Close the scissors. Store the shared prototype as one item.

32-35 min | Exit check

Students: Name your role contribution. Cite one stand test result.

What to collect or hear

One successful move and how it serves the need.

Success check

Demonstrate one successful move and explain how the design serves the stated need.

Support and stretch

Support: Grip one test object.

Stretch: Move the object past a marked line safely.

Another way to do it

Draw the mechanism and act out the move.

Optional next challenge

Gentle Grip: Modify the gripper to move a paper cup without crushing or dropping it.

Cleanup

Release tension, collect fasteners, and store test objects in their marked locations.

Teacher notes

Assistive design begins with what a person wants to do, not with assumptions about the person.

A useful gripper balances reach, control, force, and comfort.

Model respectful need-first language and a simple linkage gripper.

Demonstrate testing with lightweight objects only.

Check every moving joint and handle before use.

Assign the plan, parts, build, test, fix, and record role cards before work starts.

L26 | Tank 9: Improve one feature

GOAL: I can use test evidence to change one feature and check whether it helps.

35 minutes. Lesson reference: year:tank-revision

Before students arrive

Six table trays stocked, 30 scissors counted

Previous test records in folders

Materials

Clean cardboard; Paper; Pencils; Rulers; Student scissors; Masking tape; Reusable trays; Prototype; Previous test record; Approved test objects

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can use test evidence to change one feature and check whether it helps.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Compare a design change with a decorative change. Trace one failed test to the part causing it.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Choose one problem from your test record. Change the relevant shape, connection, or movement. Repeat the same three-trial test.

Teacher: Pause each group early. Ask what its next useful action will be.

25-30 min | Check one thing

Students: Compare the old and new results. Keep the improvement or explain why you will undo it. Record what the evidence suggests next.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show one change you made after testing. Compare the results before and after.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Before and after results for one change.

Success check

Show one change you made after testing. Compare the results before and after.

Support and stretch

Support: Pick from two suggested changes.

Stretch: Predict the result before retesting.

Another way to do it

Draw the change and explain the result you expect.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Compare the measurements and explain a tradeoff or remaining limitation.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L27 | Upcycle Lab

GOAL: Transform a clean discarded item into a useful prototype with a clearly defined user and function.

35 minutes. Lesson reference: catalog:25

Before students arrive

Clean, inspected reclaimed objects

Six table trays stocked, 30 scissors counted

Materials

Clean reclaimed objects; Cardboard; Tape; String; Fasteners; Markers; Safety scissors

Safety

Use only inspected clean items with no sharp edges, residue, glass, batteries, or unknown contents.

Run the period

00-03 min | Get ready

Students: Choose one inspected reclaimed object. Place it on the project mat.

Teacher: Upcycling adds usefulness, not just decoration. Start with the user's need, then use the object's existing strengths.

03-06 min | Material safety

Students: Use only the approved clean object. Cut only on the table. Keep fingers outside the cutting path. Carry scissors closed.

Teacher: Inspect and approve all reclaimed materials before distribution.

06-11 min | Choose a new use

Students: List two useful object properties. Name one user. Write one new function.

Teacher: Model identifying an object's useful shape, strength, or connection.

11-19 min | Build the new use

Students: Sketch the planned change. Cut only approved added material. Build the new function.

Teacher: Demonstrate a reversible prototype connection before permanent work.

19-25 min | Test the function

Students: Run one clear use test. Record the result. Find one weak feature.

25-30 min | Improve the use

Students: Change the weak feature. Repeat the use test. Prepare a before-and-after explanation.

30-32 min | Reset the lab

Students: Close the scissors. Sort every unused material.

32-35 min | Exit check

Students: Name the reused property. State the new function.

What to collect or hear

Your object's new job, explained before and after.

Success check

Demonstrate the new function and explain why it is more than decoration.

Support and stretch

Support: Give the object one new job.

Stretch: Explain why it is more than decoration.

Another way to do it

Sketch the new use and explain it.

Optional next challenge

Second Life Label: Create a simple label showing the old purpose, new purpose, and materials kept out of waste.

Cleanup

Return unused reclaimed items by material type and place true waste in the correct container.

Teacher notes

Upcycling adds usefulness, not just decoration.

Start with the user's need, then use the object's existing strengths.

Inspect and approve all reclaimed materials before distribution.

Model identifying an object's useful shape, strength, or connection.

Demonstrate a reversible prototype connection before permanent work.

L28 | Tank 10: Let someone try

GOAL: I can watch a user try my idea and learn without coaching every move.

35 minutes. Lesson reference: year:tank-user-test

Before students arrive

Reusable feedback cards out

Scrap materials in trays

Materials

Prototype; Paper; Pencils; Reusable feedback cards; Scrap materials

Safety

Use the approved classroom test. Do not test strength, comfort, or function on a person's body.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can watch a user try my idea and learn without coaching every move.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch a user receive one short instruction. Notice how the designer listens without taking over.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Tell an approved partner the product's job. Let the partner try it using the approved test. Record one clear action, one confusion, and one useful comment.

Teacher: Pause each group early. Ask what its next useful action will be.

25-30 min | Check one thing

Students: Ask what was easiest and hardest. Choose one change that would help the user. Try a quick paper or scrap version of that change.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Explain one user difficulty and the change you will make.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

One user difficulty and the change you plan.

Success check

Explain one user difficulty and the change you will make.

Support and stretch

Support: Use the watch-only checklist.

Stretch: Write down the user's exact words.

Another way to do it

A partner acts as the user with a scenario card.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Separate observed behavior from opinions and justify the selected change.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L29 | Accessibility Design Sprint

GOAL: Design and test a small prototype that reduces a specific barrier involving reach, grip, vision, hearing, movement, or attention.

35 minutes. Lesson reference: catalog:22

Before students arrive

Six table trays stocked, 30 scissors counted

Accessibility user cards and feedback cards

Materials

Accessibility user cards; Paper; Cardboard; Tape; Craft materials; Large-print labels; Feedback cards; Timer; Scissors; Access-test cards

Safety

Simulate barriers only through the provided user card. Never restrain, blindfold, label, or imitate a disability, and stop any unsafe user test.

Run the period

00-03 min | Get ready

Students: Choose one provided user card. Read the user goal aloud.

Teacher: Accessibility design starts with a person's goal and the barrier created by the environment or tool. The sprint clock keeps the team focused on access, testing, and revision instead of decoration. Present respectful user cards for one-handed use, limited reach, reduced grip, low vision, or noisy surroundings.

03-07 min | Design safely

Students: Use only the facts on the user card. Cut only on the table. Keep fingers outside the cutting path. Carry scissors closed.

07-12 min | Name the problem

Students: Name the tool or space problem. Sketch three possible changes. Circle one idea.

Teacher: Model turning a user goal and barrier into one measurable access test.

12-20 min | Build one feature

Students: Cut only the planned shapes. Build the smallest useful feature. Add a large clear label.

Teacher: Time-box ask, imagine, plan, create, test, and improve, then model a one-minute feedback interview.

20-26 min | Run the card test

Students: Use the provided test card. Observe one access task. Record where the task stops.

26-30 min | Fix the problem

Students: Change one blocked feature. Repeat the same card test.

30-32 min | Reset the sprint

Students: Close the scissors. Store the prototype with its card.

32-35 min | Exit check

Students: Write what became easier. Write what remains difficult.

What to collect or hear

A 30 second pitch: user goal, barrier, test, result, change.

Success check

Deliver a 30-second pitch that names the user goal, barrier, access test, result, and revision.

Support and stretch

Support: Pick one barrier from a user card.

Stretch: Test with a second user and revise.

Another way to do it

Draw the prototype and pitch from the drawing.

Optional next challenge

Second User: Test with a different user and identify one need the first test did not reveal.

Cleanup

Label the prototype, return unused materials, and save only the evidence needed for the pitch.

Teacher notes

Accessibility design starts with a person's goal and the barrier created by the environment or tool.

The sprint clock keeps the team focused on access, testing, and revision instead of decoration.

Present respectful user cards for one-handed use, limited reach, reduced grip, low vision, or noisy surroundings.

Model turning a user goal and barrier into one measurable access test.

Time-box ask, imagine, plan, create, test, and improve, then model a one-minute feedback interview.

L30 | Tank 11: Use materials wisely

GOAL: I can choose materials that do the job while reducing unnecessary waste.

35 minutes. Lesson reference: year:tank-materials

Before students arrive

Six table trays stocked, 30 scissors counted

Safe material samples, and a scale if available

Materials

Clean cardboard; Paper; Pencils; Rulers; Student scissors; Masking tape; Reusable trays; Safe material samples; Prototype; Shared scale if available

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can choose materials that do the job while reducing unnecessary waste.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Compare two materials doing the same small job. Notice what can be reused or made smaller.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Sort the design's parts into essential and optional. Compare two suitable materials for one part. Improve that part without making the product harder to use.

Teacher: Pause each group early. Ask what its next useful action will be. Use qualitative comparisons if no scale exists. Do not invent purchase costs or environmental claims.

25-30 min | Check one thing

Students: Check that the main job still works. Count pieces or compare material size before and after. Sort useful scraps for another group.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show a useful material choice and explain what you saved or improved.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

One part made smaller or reused, noted in the folder.

Success check

Show a useful material choice and explain what you saved or improved.

Support and stretch

Support: Swap one material.

Stretch: Estimate how much material you saved.

Another way to do it

Compare materials by touch and weight only.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Explain the tradeoff among performance, quantity, reuse, and finish.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L31 | Moving Picture Machine

GOAL: Create a sequence device that uses small image changes to produce the illusion of motion.

35 minutes. Lesson reference: catalog:18

Before students arrive

Index cards, strips and fasteners in trays

Materials

Index cards; Paper strips; Cardstock circles; Pencils; Markers; Brass fasteners; Binder clips

Safety

Keep spinning devices at table level and handle fasteners only as demonstrated.

Run the period

00-03 min | Get ready

Students: Choose flip, wheel, or strip motion. Place the frame paper in order.

Teacher: Motion can emerge when the brain sees a rapid sequence of slightly different images. Consistent alignment makes each change easier to see.

03-06 min | Fastener safety

Students: Keep the device at table level. Use fasteners only through marked holes. Ask the teacher about a sharp fastener.

Teacher: Model a two-frame flip image and a multi-frame sequence.

06-10 min | Plan the motion

Students: Draw a clear start pose. Draw a clear finish pose. Mark the matching dot.

Teacher: Show registration marks that keep frames aligned.

10-19 min | Draw the frames

Students: Change one small part per frame. Keep each drawing on the registration mark. Number the frames.

Teacher: Demonstrate a safe spinning or flipping motion.

19-26 min | Assemble the motion

Students: Stack or attach the frames in order. Secure the approved fastener. Move the device slowly.

26-30 min | Smooth the motion

Students: Find the largest visual jump. Revise one frame. Test at two speeds.

30-32 min | Sort the pieces

Students: Return unused fasteners. Clip your frames together.

32-35 min | Exit check

Students: Show the finished motion once. Name the frame you changed.

What to collect or hear

A running animation with one smoothed frame.

Success check

Run the animation and identify one frame changed to make the motion smoother.

Support and stretch

Support: Make four frames with small changes.

Stretch: Smooth one frame and explain the change.

Another way to do it

A paper flip book of the same frames.

Optional next challenge

Smooth Loop: Add transition frames so the final image flows naturally back to the first.

Cleanup

Cap markers, collect fasteners, and store finished sequences flat.

Teacher notes

Motion can emerge when the brain sees a rapid sequence of slightly different images.

Consistent alignment makes each change easier to see.

Model a two-frame flip image and a multi-frame sequence.

Show registration marks that keep frames aligned.

Demonstrate a safe spinning or flipping motion.

L32 | Tank 12: Name and package

GOAL: I can communicate my product's purpose with a clear name, logo, and simple package.

35 minutes. Lesson reference: year:tank-brand-package

Before students arrive

Markers, reclaimed boxes and tape out

Materials

Prototype; Paper; Shared markers; Reclaimed boxes; Cardboard; Tape; Scissors; Ruler

Safety

Use the approved gentle handling test. No throwing or heated forming without a separate approved teacher station.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can communicate my product's purpose with a clear name, logo, and simple package.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Compare a readable product label with a crowded one. Watch a package protect the model and open easily.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Choose a clear product name and reuse your logo skills. Make a label that tells the product's job. Create or improve a simple protective package.

Teacher: Pause each group early. Ask what its next useful action will be.

25-30 min | Check one thing

Students: Ask a partner to read the label without help. Check opening, fit, and the approved handling test. Revise one confusing or wasteful detail.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Show a readable product label and a package that opens and protects.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

A readable label and a protective package.

Success check

Show a readable product label and a package that opens and protects.

Support and stretch

Support: A name and one symbol.

Stretch: Explain how the label fits your user.

Another way to do it

Make the label on paper only.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Connect branding and packaging choices to the user and test evidence.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L33 | Fix, Remix, or Invent Studio

GOAL: Choose to repair a flawed prototype, remix an earlier challenge with a new constraint, or invent a small solution from a prompt, then test it with evidence.

35 minutes. Lesson reference: catalog:34

Before students arrive

Class crate out, labeled with the class code only

Repair, remix and invention cards

Everyday maker bin

Materials

Repair cards and safe flawed prototypes; Remix constraint cards; Invention prompt cards; Everyday Maker Bin; Test tools; Before-and-after record; Safety tools as assigned; Path safety cards

Safety

Follow the safety card for the selected path, inspect before testing, and stop any prototype with an unexpected hazard.

Run the period

00-03 min | Get ready

Students: Choose fix, remix, or invent. Read the matching safety card.

Teacher: Student Studio means the team chooses a path, but every path still needs a clear test and evidence. Fix, remix, or invent, then change one meaningful thing at a time so the result can teach us something.

03-07 min | Studio safety

Students: Inspect every material before use. Cut only on the table. Keep fingers outside the cutting path. Carry scissors closed. Stop any unexpected hazard.

Teacher: Set three clearly labeled stations with repair cards, remix constraints, and invention prompts.

07-11 min | Set the test

Students: Write the prompt. Write the rule you must follow. Define one success test.

Teacher: Model a baseline or first test for each path and require a success criterion before materials are collected.

11-19 min | Run the first test

Students: Build or inspect the first version. Run the success test. Record the most useful evidence.

Teacher: Use short team conferences instead of continuous whole-class directions.

19-25 min | Make one change

Students: Choose one change because of the test result. Cut only the planned material. Complete the change.

25-30 min | Retest the change

Students: Repeat the same success test. Keep, revise, or reject the change. Record your reason.

30-32 min | Reset the studio

Students: Close the scissors. Sort all reusable parts.

32-35 min | Exit check

Students: Compare the two test results. Name your next change.

What to collect or hear

Your chosen path, a baseline test and a first attempt.

Success check

Name the chosen path, show before-and-after evidence, and defend whether the change should be kept.

Support and stretch

Support: Pick a repair card with a clear flaw.

Stretch: Invent from a prompt and set your own success test.

Another way to do it

Plan the fix, remix or invention on paper.

Optional next challenge

Studio Guide: Write one concise tip that helps the next team start the chosen path without giving away the solution.

Cleanup

Return each station to its start card, sort materials, label work selected for later use, and save evidence with the matching path.

Teacher notes

Student Studio means the team chooses a path, but every path still needs a clear test and evidence.

Fix, remix, or invent, then change one meaningful thing at a time so the result can teach us something.

Set three clearly labeled stations with repair cards, remix constraints, and invention prompts.

Model a baseline or first test for each path and require a success criterion before materials are collected.

Use short team conferences instead of continuous whole-class directions.

L34 | Tank 13: Explain your idea

GOAL: I can explain the problem, my solution, and evidence that helped me improve it.

35 minutes. Lesson reference: year:tank-pitch

Before students arrive

Index cards and test records out

Materials

Prototype; Previous test record; Index cards; Pencils; Product label

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can explain the problem, my solution, and evidence that helped me improve it.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Hear a short pitch with a problem, idea, proof, and next step. Notice a speaker showing the product while explaining it.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Make four short cue cards or picture prompts. Choose one clear test result as proof. Prepare a simple demonstration of the product.

Teacher: Pause each group early. Ask what its next useful action will be. No sales claims or money request is needed. A classroom pitch explains design decisions.

25-30 min | Check one thing

Students: Tell the story to one partner. Ask the partner to repeat the main idea. Cut details that hide the problem, solution, or proof.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Give a short practice pitch that includes the problem, solution, and one piece of evidence.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Four cue cards and one practice pitch.

Success check

Give a short practice pitch that includes the problem, solution, and one piece of evidence.

Support and stretch

Support: Use a four-line pitch frame.

Stretch: Add one number from your testing.

Another way to do it

Pitch with folder pages, or hand in a written card.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Use a roughly 60-second explanation with evidence, a design tradeoff, and a realistic next step.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L35 | Innovation Studio 2: Change one thing and retest

GOAL: I can change one part of my design, run the same test, and decide to keep or undo it.

35 minutes. Lesson reference: year:studio-retest

Before students arrive

Class crate out, labeled with the class code only

Leftover maker bin and small trays out

Materials

Your studio model from last time; Before and after record sheet; Leftover maker bin; Small tray per team; Test tools from last time

Safety

Use teacher-approved materials only. The teacher manages any knife cuts or hot glue. Keep the test the same as last time.

Run the period

00-02 min | Get ready

Students: Get your model and record sheet. Read the goal: one change, same test.

Teacher: Hand out models from the class crate.

02-03 min | Find your baseline

Students: Find last time's test result.

Teacher: Check every team has a baseline number or note.

03-07 min | Read your baseline

Students: Look at last time's test result. Pick the one part that held you back.

Teacher: Ask: which part failed first?

07-19 min | Make one change

Students: Change one part only. Keep the test the same.

Teacher: Open the teacher station for cuts or glue at set times.

19-27 min | Retest

Students: Run the same test again. Record the new result next to the old one.

Teacher: Watch that the test really is the same.

27-30 min | Keep or undo

Students: Did the change help? Keep it or undo it. Write why.

Teacher: Accept keep or undo. Both are good engineering.

30-33 min | Reset together

Students: Return tools and reusable pieces. Put unfinished work in the class tray. Clear your table and floor.

Teacher: Call tool returns first. Count scissors back before the line forms.

33-35 min | Show your result

Students: Show your before and after results to the teacher.

Teacher: Look for two results side by side.

What to collect or hear

Before and after results and a keep or undo reason.

Success check

Show your before and after results to the teacher.

Support and stretch

Support: Pick one of two change cards.

Stretch: Predict the result before retesting and explain any surprise.

Another way to do it

Draw the change, act out the test, then decide keep or undo.

Optional next challenge

Write the next change you would try and why.

Cleanup

Return tools and reusable pieces.

Put unfinished work in the class tray.

Clear your table and floor.

Teacher notes

Grade 5 only, second studio meeting. It continues the Fix, Remix, or Invent Studio from last time.

More challenge: ask the team to predict the result before retesting and explain any surprise.

No-tech path: teams draw the change and act out the test, then decide keep or undo with reasons.

L36 | Tank 14: Rehearse and refine

GOAL: I can practice a clear presentation and improve it after specific feedback.

35 minutes. Lesson reference: year:tank-rehearse

Before students arrive

Timer on the screen

Feedback cards out

Materials

Prototype; Cue cards; Existing classroom timer; Reusable feedback cards; Display surface

Safety

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can practice a clear presentation and improve it after specific feedback.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch how speakers take turns and show their model on the table. Compare useful feedback with a vague compliment.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Choose who will explain or demonstrate each part. Rehearse once while a partner times the explanation. Practice one likely question and an honest answer.

Teacher: Pause each group early. Ask what its next useful action will be. Students may point, demonstrate, or use a partner. Speaking alone is not the only way to participate.

25-30 min | Check one thing

Students: Ask for one strength and one specific improvement. Change the unclear part. Rehearse again with a reliable reset.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Deliver your revised pitch and show how the product resets for the next visitor.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

A revised, timed pitch.

Success check

Deliver your revised pitch and show how the product resets for the next visitor.

Support and stretch

Support: Practice once with a partner, then once for the class.

Stretch: Cut your pitch to the time limit and keep the evidence.

Another way to do it

Rehearse with a partner using written cues.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Stay near the agreed time and answer a question using evidence rather than guesses.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L37 | Innovation Share

GOAL: I can show my problem, process, evidence and next step to my classmates.

35 minutes. Lesson reference: year:innovation-share

Before students arrive

Class crate out, labeled with the class code only

Four label cards per team

Materials

Studio models; Team folders with KidWind and challenge records; Label cards; Existing display surfaces; Pencils

Safety

Keep walkways clear during the gallery walk. Touch other models only when invited.

Run the period

00-02 min | Get ready

Students: Get your model and folder. Read the goal: show four parts.

Teacher: Show the four label cards.

02-03 min | Find your work

Students: Find one record from KidWind or a challenge.

Teacher: Help teams find records in folders.

03-07 min | Watch an exhibit

Students: See the four parts: problem, process, evidence, next step.

Teacher: Model one exhibit quickly with a sample.

07-19 min | Build your exhibit

Students: Set out your model. Add one record as evidence. Write four short labels.

Teacher: Check each exhibit has all four labels.

19-29 min | Gallery walk

Students: Visit two exhibits. Leave one kind, specific comment.

Teacher: Rotate groups on a signal.

29-33 min | Take home decision

Students: Decide: take home, class sample, or recycle. Empty your crate cell.

Teacher: Crates should be empty by the end of today.

33-35 min | Reflect

Students: Name one thing you would change next time.

Teacher: Accept a spoken or written answer.

What to collect or hear

A four-label exhibit, then a take-home decision.

Success check

Name one thing you would change next time.

Support and stretch

Support: Use the sentence starters on the label cards.

Stretch: Add a chart or number to the evidence label.

Another way to do it

Share by pointing to labels or with a written card.

Optional next challenge

Write a kind, specific comment for one more exhibit.

Cleanup

Decide: take home, class sample, or recycle.

Empty your crate cell.

Teacher notes

Grade 5 only. Adapted from the CIRC Showcase Builder for one in-class period. No outside visitors are assumed.

More challenge: add one chart or number to the evidence label.

No-tech or quiet path: a student can share by pointing to labels or by a written card instead of talking.

L38 | Tank 15: Share and reflect

GOAL: I can show my design, listen to feedback, and explain how it changed.

35 minutes. Lesson reference: year:tank-showcase

Before students arrive

Display surfaces cleared

Labels and first sketches out

Materials

Prototype; First sketch; Test evidence; Labels; Existing display surfaces; Paper; Pencils

Safety

Keep demonstrations at the approved stations. Reset loose parts and walk between displays.

Run the period

00-02 min | Get ready

Students: Read today's goal. Look at the example.

Teacher: Say today's goal in plain words: I can show my design, listen to feedback, and explain how it changed.

02-03 min | Tell your partner

Students: Tell your partner one useful detail.

Teacher: Listen for one useful observation. Connect it to today's demonstration.

03-08 min | Watch and try

Students: Watch a visitor approach, hear a short pitch, and ask one question. Notice how the presenter resets the space.

Teacher: Model the demonstration actions with one example. Show the unfinished stage, not only a polished result.

08-25 min | Make it work

Students: Set out the prototype, label, and one piece of test evidence. Take turns presenting and visiting other teams. Give one useful observation and one respectful question.

Teacher: Pause each group early. Ask what its next useful action will be. This is an in-class share unless a separate audience or event has been approved. Do not promise outside visitors.

25-30 min | Check one thing

Students: Compare your first sketch with your final version. Choose the most useful change. Name one thing you would test next.

Teacher: Ask for the evidence named in this check. Give one specific next step.

30-33 min | Reset together

Students: Return tools and reusable pieces. Label your work with the class storage code. Clear your table and floor.

Teacher: Call tool returns first. Check table trays and keep unfinished work for the next contact.

33-35 min | Show your result

Students: Explain how evidence changed your design and identify a sensible next test.

Teacher: Listen or look for the stated result. Accept a spoken explanation or a pointing demonstration.

What to collect or hear

Your first sketch next to your final model, with evidence. Then decide what goes home.

Success check

Explain how evidence changed your design and identify a sensible next test.

Support and stretch

Support: Show your model and one piece of evidence.

Stretch: Explain how the evidence changed your design.

Another way to do it

Share by pointing to labels or with a written card.

Optional next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Cleanup

Return tools and reusable pieces.

Label your work with the class storage code.

Clear your table and floor.

Teacher notes

Grade 6 CIRC Tank: Explain a supported improvement, remaining limitation, and next test.

Teacher-led classroom work. No new student account, grade, or online assignment is required.

Keep student names off public screens. Use class storage codes for shared work.

L39 | Hydroponics Flow Lab

GOAL: Model a gravity-fed hydroponic path that distributes water evenly without leaking or flooding.

35 minutes. Lesson reference: catalog:26

Before students arrive

Water trays, towels and measuring cups

Tape marks the wet zone. Laptops away.

Materials

Clean tubing; Cups; Reclaimed channels; Water trays; Measuring cups; Towels; Plant-site models; Teacher-controlled tubing cutter; Wet mats; Sink bucket

Safety

Keep water away from outlets and electronics, wipe spills immediately, and use no pump unless the teacher controls it.

Run the period

00-03 min | Get ready

Students: Place the dry tray on the wet mat. Measure the test water.

Teacher: Plants need access to water and air, so more water is not automatically better. A good flow model makes distribution visible and measurable.

03-07 min | Wet tool safety

Students: Keep water inside the tray. Wipe each spill with a towel. Do not cut tubing yourself. Keep electronics away from the wet zone.

Teacher: Model a dry layout before adding any water.

07-10 min | Plan the flow

Students: Draw the water cup. Draw the water path. Mark each plant spot and the end cup.

Teacher: Demonstrate a measured pour and leak check at every connection.

10-17 min | Build the dry model

Students: Mark the cut line on the tube. Give the tube to the teacher. Wait for the cut, then connect the tube path. Check every connection while dry.

Teacher: Keep all water testing inside the protected wet zone.

17-23 min | Run the water

Students: Pour the measured water into the water cup. Watch each plant spot. Measure the collected water.

23-29 min | Change the flow

Students: Change one slope, hole, or loose connection. Repeat with the same water amount. Record the new collected amount.

29-33 min | Dry the lab

Students: Drain the tray into the sink bucket. Place wet parts on the drying tray. Dry the table and floor.

33-35 min | Exit check

Students: Write the collected amount. Name the flow change.

What to collect or hear

A measured flow result and one change that made the flow more even.

Success check

Show the measured flow result and explain the one change that made distribution more even.

Support and stretch

Grade 5: Measure water at each plant site with a cup. Mark more or less.

Grade 6: Calculate the percent of water recovered and name where it was lost.

Another way to do it

A dry path model with one teacher pour demo.

Optional next challenge

Flow Alarm: Design a visible marker that shows when one plant site receives too much or too little water.

Cleanup

Drain models into the assigned container, dry all surfaces, and separate wet materials for drying.

Teacher notes

Plants need access to water and air, so more water is not automatically better.

A good flow model makes distribution visible and measurable.

Model a dry layout before adding any water.

Demonstrate a measured pour and leak check at every connection.

Keep all water testing inside the protected wet zone.

L40 | Tinkercad 1: Move, turn, size

GOAL: I can place shapes in Tinkercad and control their size, height and turn.

35 minutes. Lesson reference: year:tinker-move

Before students arrive

Laptop cart charged and counted

Tinkercad class code on the board, nicknames only

Materials

Cart laptops, charged; Tinkercad Classroom class code, nicknames only; Projected teacher demo; Grid paper and rulers for the no-device path

Safety

Join with the class code and a nickname only. No personal email. Use initials, not full names, on anything shown or printed.

Run the period

00-04 min | Sign in

Students: Get your laptop. Join the class with the code and your nickname.

Teacher: Write the class code on the board. Nobody signs in for a student.

04-06 min | Tell your partner

Students: Tell your partner something you have seen that was 3D printed.

Teacher: Take one or two answers.

06-10 min | Watch the handles

Students: Watch the teacher drag a box. See how to size, lift and turn it.

Teacher: Show the white handles, the black lift arrow and the curved turn arrow.

10-24 min | Make a name plate

Students: Drag a box for the base. Add letters with your initials or nickname. Size the letters to fit the base.

Teacher: Circulate. Ask students to show you the side view.

24-28 min | Check it

Students: Look from the side. Letters sit on the base, not floating or sunk. Show a partner your side view on screen.

Teacher: Fix one floating plate with the class watching.

28-32 min | Save and shut

Students: Check that your work saved. Close the lid with two hands. Put the laptop back in its slot and plug it in.

Teacher: Count the cart before the class leaves.

32-35 min | Tell how

Students: Tell a partner how you kept the letters on the base.

Teacher: Accept a spoken answer. Laptops are already put away.

What to collect or hear

A name plate with letters sitting on the base, shown on screen.

Success check

Tell a partner how you kept the letters on the base.

Support and stretch

Grade 5: Follow each step as written.

Grade 6: Make the base a different shape.

Another way to do it

Draw the name plate to millimeter size on grid paper and cut it from card.

Optional next challenge

Make the base a different shape and keep the letters sitting on it.

Cleanup

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Teacher notes

Confirm Tinkercad approval and teacher access before this unit. If access is not approved, run the grid paper version of the same six visits.

No-device path: draw a name plate to exact millimeter size on grid paper and cut it from card.

L41 | Tinkercad 2: Holes and group

GOAL: I can use a hole shape and group to make a keychain tag with a working hole.

35 minutes. Lesson reference: year:tinker-holes

Before students arrive

Laptop cart charged and counted

Tinkercad class code on the board

Materials

Cart laptops, charged; Tinkercad Classroom class code; Projected teacher demo; Card, hole punch and grid paper for the no-device path

Safety

Nicknames or initials only. Laptops stay on the table.

Run the period

00-04 min | Sign in

Students: Get your laptop. Open Tinkercad with the class code.

Teacher: Help one table at a time.

04-05 min | Recap

Students: Show a partner how to lift a shape.

Teacher: Listen for the black lift arrow.

05-09 min | Watch a hole

Students: Watch a cylinder turn into a hole. Watch Group cut it out.

Teacher: Show the preview from below so the hole is clear.

09-23 min | Make a tag

Students: Make a box for your keychain tag. Set a cylinder to Hole and push it through a corner. Select both and choose Group.

Teacher: Ask: can you see through it yet?

23-28 min | Check the hole

Students: Can you see through the hole? If not, push it all the way through. Show a partner the hole from underneath.

Teacher: Fix one hole that stops halfway.

28-32 min | Save and shut

Students: Check that your work saved. Close the lid with two hands. Put the laptop back in its slot and plug it in.

Teacher: Count the cart before the class leaves.

32-35 min | Tell how

Students: Tell a partner how you made the hole go all the way through.

Teacher: Accept a spoken answer. Laptops are already put away.

What to collect or hear

A keychain tag with a hole you can see through.

Success check

Tell a partner how you made the hole go all the way through.

Support and stretch

Grade 5: Make the tag and the working hole.

Grade 6: Add a raised letter grouped with the tag.

Another way to do it

Draw the tag to size, punch a clean hole and join two card layers.

Optional next challenge

Add a raised letter and group it with the tag.

Cleanup

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Teacher notes

Second of six Tinkercad visits.

No-device path: draw the tag to size, punch a clean hole, and join two card layers.

L42 | Tinkercad 3: Align and mirror

GOAL: I can use duplicate, mirror and align to make both sides of a model match.

35 minutes. Lesson reference: year:tinker-mirror

Before students arrive

Laptop cart charged and counted

Tinkercad class code on the board

Materials

Cart laptops, charged; Tinkercad Classroom class code; Projected teacher demo; Folded grid paper for the no-device path

Safety

Nicknames or initials only. Laptops stay on the table.

Run the period

00-04 min | Sign in

Students: Get your laptop. Open Tinkercad with the class code.

Teacher: Help one table at a time.

04-05 min | Recap

Students: Tell a partner what Group does.

Teacher: Listen for joining shapes into one.

05-09 min | Watch mirror

Students: Watch one half become two with Mirror. Watch Align line them up.

Teacher: Show duplicate, mirror, then align, in that order.

09-24 min | Build a matching model

Students: Build one half of a trophy or rocket. Duplicate it, then Mirror it. Use Align to line both halves up.

Teacher: Ask students to explain which way they mirrored.

24-28 min | Check both sides

Students: Do both sides match? Fix one side if they do not. Show a partner both sides on screen.

Teacher: Check from the front view.

28-32 min | Save and shut

Students: Check that your work saved. Close the lid with two hands. Put the laptop back in its slot and plug it in.

Teacher: Count the cart before the class leaves.

32-35 min | Name the tool

Students: Name the tool that helped your two sides match.

Teacher: Accept a spoken answer. Laptops are already put away.

What to collect or hear

A model with matching sides.

Success check

Name the tool that helped your two sides match.

Support and stretch

Grade 5: Build one half, then mirror and align it.

Grade 6: Add initials that stay readable after mirroring.

Another way to do it

Draw one half on folded grid paper, cut, and open it.

Optional next challenge

Add your initials and check that they read the right way.

Cleanup

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Teacher notes

Third of six Tinkercad visits.

No-device path: draw one half on folded grid paper, cut, and open to check both sides match.

L43 | Tinkercad 4: Design a skate park piece

GOAL: I can design a skate park piece that fits the size rule and has a flat bottom.

35 minutes. Lesson reference: year:tinker-ramp

Before students arrive

Laptop cart charged and counted

Size rule card: 80 by 60 by 40 millimeters

One printed test ramp if a printer is ready

Materials

Cart laptops, charged; Tinkercad Classroom class code; One printed test ramp if a printer is ready; Test boards or finger boards if approved, cardboard boards otherwise; Rulers and size rule card: 80 by 60 by 40 millimeters

Safety

Nicknames or initials only. Test boards stay on tables.

Run the period

00-04 min | Sign in

Students: Get your laptop. Open Tinkercad with the class code.

Teacher: Hold up the test ramp.

04-05 min | Measure a board

Students: Hold a test board. How wide is it?

Teacher: A finger board is about 26 millimeters wide.

05-09 min | Watch the size rule

Students: No piece bigger than 80 by 60 by 40 millimeters. Flat bottom, at least 40 millimeters wide.

Teacher: Show where Tinkercad shows the size numbers.

09-24 min | Design your piece

Students: Choose a kicker ramp, quarter pipe, grind box or rail. Start from a wedge or a box. Check your size with the numbers.

Teacher: Circulate with the size rule card.

24-28 min | Teacher check

Students: Show the flat bottom and the size. Fix anything flagged.

Teacher: Flag size and flat bottom only. Write a short note.

28-32 min | Save and shut

Students: Check that your work saved. Close the lid with two hands. Put the laptop back in its slot and plug it in.

Teacher: Count the cart before the class leaves.

32-35 min | Show your result

Students: Tell a partner which piece you made and its size.

Teacher: Accept a spoken answer.

What to collect or hear

A skate park piece that passes the size and flat bottom check.

Success check

Tell a partner which piece you made and its size.

Support and stretch

Grade 5: Pick one piece and meet the size rule.

Grade 6: Add a grip texture or logo and still meet the size rule.

Another way to do it

Draw the piece on millimeter grid paper at 1:1 and build it in thin cardboard.

Optional next challenge

Add a grip texture or a small logo without breaking the size rule.

Cleanup

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Teacher notes

Fourth of six Tinkercad visits. Print and measure one test ramp before this visit if a printer is available.

No-device path: draw the piece on millimeter grid paper at 1:1 and build it in thin cardboard.

L44 | Tinkercad 5: Fix, export, queue

GOAL: I can fix my design, export it, and put it in the class print folder with a clear name.

35 minutes. Lesson reference: year:tinker-export

Before students arrive

Laptop cart charged and counted

Class print folder set up

File name pattern on the board

Materials

Cart laptops, charged; Tinkercad Classroom class code; Class print folder the teacher set up; File name pattern on the board: table number and piece

Safety

Nicknames or initials only in file names.

Run the period

00-04 min | Sign in

Students: Get your laptop. Open your skate park piece.

Teacher: Hand back teacher notes from last time.

04-05 min | Read your note

Students: Read the teacher note about your design.

Teacher: Answer questions about notes quickly.

05-17 min | Fix flagged parts

Students: Fix what the note says. Check the size and flat bottom again.

Teacher: Check fixes before export.

17-24 min | Export

Students: Choose Export, then STL. Name the file with your table number and piece.

Teacher: Show the export button on the projector.

24-28 min | Queue it

Students: Put your file in the class print folder. Check that your file name is there. Remember your file label and folder.

Teacher: Watch the folder fill. Help anyone missing.

28-32 min | Save and shut

Students: Check that your work saved. Close the lid with two hands. Put the laptop back in its slot and plug it in.

Teacher: Count the cart before the class leaves.

32-35 min | Tell the teacher

Students: Tell the teacher your file label and where you saved it.

Teacher: Check the folder yourself later. Laptops are already put away.

What to collect or hear

An STL file in the class print folder, named with your table number.

Success check

Tell the teacher your file label and where you saved it.

Support and stretch

Grade 5: Fix and export one piece.

Grade 6: Design a second piece that connects to the first.

Another way to do it

Fix the paper drawing and cut the final cardboard piece at 1:1.

Optional next challenge

Design a second piece that connects to your first.

Cleanup

Check that your work saved.

Close the lid with two hands.

Put the laptop back in its slot and plug it in.

Teacher notes

Fifth of six Tinkercad visits. Printing happens between this visit and the next.

Print cap: measure one print first, then print in queue order up to what the printer can finish. One piece per team is a simple rule.

No-device path: fix the paper drawing and cut the final cardboard piece at 1:1.

L45 | Tinkercad 6: Build day and pitch

GOAL: I can assemble our table's skate park, test it, and pitch what we made.

35 minutes. Lesson reference: year:tinker-pitch

Before students arrive

Printed pieces laid out by table

Thin cardboard and tape for unprinted pieces

Park base sheet per table

Test boards

Materials

Printed pieces so far, laid out by table; Thin cardboard and tape for any piece not printed; Size gauge; Park base sheet per table; Test boards

Safety

Pieces stay on the tables. Cardboard cuts use student scissors only.

Run the period

00-02 min | Get ready

Students: Find your table's pieces. Read the goal: build, test, pitch.

Teacher: Lay out printed pieces by table before class.

02-03 min | Plan the park

Students: Decide where each piece goes.

Teacher: Hand out base sheets.

03-13 min | Build the park

Students: Place your table's pieces on the base sheet. Use cardboard for any piece not printed.

Teacher: Help cardboard builders match the numbers.

13-19 min | Test with a board

Students: Roll a board over each piece. Fix one thing that catches.

Teacher: Watch for pieces that tip.

19-27 min | Pitch

Students: Give your pitch. Say what it is, what was hard, and what you would change.

Teacher: Call tables in order and keep time.

27-32 min | Take home rule

Students: Follow the take home rule for printed pieces. Recycle cardboard scraps.

Teacher: Say the take home rule clearly before cleanup.

32-35 min | Show your result

Students: Name one skill you used from all six visits.

Teacher: Accept a spoken answer.

What to collect or hear

Your pitch, and your piece tested on the table park.

Success check

Name one skill you used from all six visits.

Support and stretch

Grade 5: Pitch in 30 to 45 seconds with a partner: what it is and what was hard.

Grade 6: Give a 60 second pitch that adds what you would change and why.

Another way to do it

Build any unprinted piece in cardboard from the same numbers.

Optional next challenge

Add one connecting piece in cardboard so two pieces flow together.

Cleanup

Follow the take home rule for printed pieces.

Recycle cardboard scraps.

Teacher notes

Last of six Tinkercad visits. Every student tests and pitches, printed or not.

Build any unprinted piece in thin cardboard at 1:1 from the same millimeter numbers.

L46 | Outdoor Microclimate Map

GOAL: Compare conditions across outdoor classroom zones and recommend where different learning activities should happen.

35 minutes. Lesson reference: catalog:6

Before students arrive

30 clipboards counted

Zone map copies

Boundaries decided and weather checked

Indoor weather data cards ready

Materials

Clipboards; Outdoor zone map; Pencils; Thermometers if available; Wind ribbons; Observation key; Indoor weather data cards

Safety

Stay inside the announced boundaries, walk between zones, do not touch unknown plants or animals, and use tools only as demonstrated.

Run the period

00-03 min | Get ready

Students: Clip the zone map to your board. Check your assigned zones.

Teacher: One outdoor classroom can contain several tiny climates at the same time. A useful map combines measurement, observation, and a recommendation for a real activity. Provide indoor data cards if weather or air quality prevents outdoor work.

03-05 min | Walk out

Students: Walk with your weather team.

05-08 min | Use tools safely

Students: Carry the thermometer by its case. Leave unknown plants untouched. Stay inside each marked zone.

Teacher: Mark the approved outdoor zones and model one sun, shade, wind, surface, and comfort observation.

08-16 min | Observe two zones

Students: Mark sun or shade. Mark the surface condition. Watch the wind ribbon.

Teacher: Assign each team two zones and demonstrate safe thermometer or wind-ribbon use when those tools are available.

16-22 min | Take readings

Students: Take one approved reading in each zone. Write each number on the map.

22-27 min | Build the map

Students: Add both zones to the class key. Choose the best zone for one activity.

27-30 min | Pack the tools

Students: Return each tool to its case. Clip all maps together.

30-32 min | Return inside

Students: Walk back with your weather team.

32-35 min | Exit check

Students: Write two observations that support your zone choice.

What to collect or hear

Two mapped zones and one activity recommendation.

Success check

Point to two mapped zones and defend one activity recommendation with evidence.

Support and stretch

Grade 5: Record sun or shade, wind, surface and comfort with symbols.

Grade 6: Add one measured reading if a tool exists. Defend a choice with two observations.

Another way to do it

Use the indoor weather data cards.

Optional next challenge

Weather Shift: Predict which mapped zone would change most after rain, strong wind, or a very sunny afternoon.

Cleanup

Return clipboards and tools by count, remove all wind ribbons, and leave every outdoor zone unchanged.

Teacher notes

One outdoor classroom can contain several tiny climates at the same time.

A useful map combines measurement, observation, and a recommendation for a real activity.

Mark the approved outdoor zones and model one sun, shade, wind, surface, and comfort observation.

Assign each team two zones and demonstrate safe thermometer or wind-ribbon use when those tools are available.

Provide indoor data cards if weather or air quality prevents outdoor work.

L47 | Reflect, reset, take home

GOAL: I can look back at my year of work, name what I learned, and take my work home.

35 minutes. Lesson reference: year:reflect-reset

Before students arrive

Student work trays out

Reflection half sheets

Inventory sheet for reusable sets

Materials

Student work and records from the year; Reflection half sheets; Pencils; Inventory sheet for reusable sets; Bags or folders for take home

Safety

Carry models with two hands. Keep walkways clear.

Run the period

00-02 min | Get ready

Students: Find your work from this year. Read the goal: look back and take it home.

Teacher: Have work trays out before class.

02-03 min | Tell your partner

Students: Tell a partner your favorite CIRC day.

Teacher: Take one or two answers.

03-11 min | Lay out your year

Students: Lay out your records, models and files. Find one piece you are proud of.

Teacher: Help students find missing pieces.

11-21 min | Write your reflection

Students: Draw or write one thing you learned. Write one thing you would change.

Teacher: Read reflections over shoulders. Ask one follow up.

21-27 min | Share one thing

Students: Share your proud piece with a partner.

Teacher: Pair students who have not talked yet.

27-33 min | Take it home

Students: Pack your work to take home. Help count the reusable sets.

Teacher: Collect the inventory sheet.

33-35 min | Show your result

Students: Hand in your reflection sheet.

Teacher: Collect sheets at the door.

What to collect or hear

Your reflection sheet. Everything else goes home.

Success check

Hand in your reflection sheet.

Support and stretch

Grade 5: Draw or write one thing you learned and one thing you would change.

Grade 6: Write one evidence-backed tip for next year's class.

Another way to do it

Draw, point to your work, or tell the teacher one sentence.

Optional next challenge

Help count one reusable set on the inventory sheet.

Cleanup

Pack your work to take home.

Help count the reusable sets.

Teacher notes

Last meeting of the year. Everything goes home or gets recycled. Reusable sets get counted.

No-tech or quiet path: students can draw, point to work, or tell the teacher one sentence.

L48 | Welcome to CIRC

GOAL: I can find my way around CIRC and help set our trust and respect rules.

35 minutes. Lesson reference: year:b-welcome

Before students arrive

Room map on the board

Chart paper and markers

Ten index cards per table

Materials

Room map on the board; Chart paper for trust and respect; Markers; Ten index cards per table for the quick build

Safety

Walk in the room. Keep bags and chairs out of walkways.

Run the period

00-03 min | Find your seat

Students: Find your seat. Read the goal on the board.

Teacher: Greet at the door. Point to seats.

03-05 min | Meet a partner

Students: Tell a partner one thing you like to make.

Teacher: Model a short answer first.

05-11 min | How CIRC works

Students: Look at our room map. Find the trays, tools and the door line.

Teacher: Walk the room map on the screen.

11-19 min | Trust and respect

Students: With your table, write what trust looks like here. Do the same for respect.

Teacher: Collect ideas onto one class chart.

19-28 min | Quick build

Students: Build the tallest tower you can with ten cards. Test it. Improve it once.

Teacher: Point out one improvement you see.

28-32 min | Reset together

Students: Return tools and reusable pieces. Put unfinished work in the class tray. Clear your table and floor.

Teacher: Call tool returns first. Count scissors back before the line forms.

32-35 min | Show your result

Students: Say one rule you will keep in CIRC.

Teacher: Accept a spoken or written answer.

What to collect or hear

Your table's trust and respect ideas on the class chart.

Success check

Say one rule you will keep in CIRC.

Support and stretch

Grade 5: Learn every tray and the door line.

Grade 6: Explain one CIRC routine to the room.

Another way to do it

Add a sticky note instead of speaking.

Optional next challenge

Improve your tower once more and explain what changed.

Cleanup

Return tools and reusable pieces.

Put unfinished work in the class tray.

Clear your table and floor.

Teacher notes

First meeting of the year. Build routines and belonging before content.

No-tech or quiet path: students can add a sticky note instead of speaking.

L49 | Set Up Your Laptop

GOAL: I can set up my school laptop so it is comfortable to use, and I can reach Teams, OneDrive and the right printer on my own.

35 minutes. Lesson reference: year:setup-devices

Before students arrive

One charged laptop per student

Projected teacher screen

Hallway printer name on the board, only if time allows

Materials

One school laptop per student, charged; Student school account sign-in; The hallway printer name on the board, only if you have time for the optional last step; A projected copy of the teacher screen

Safety

Nobody signs in as anybody else. If a student cannot get into their account, they pair with a neighbour and watch rather than borrowing a login. Laptops stay on the table, not on laps, and go back on the cart at the end.

Run the period

00-04 min | Left and right click

Students: Find the two bottom corners of the trackpad. LEFT click picks something. It is the one you use most. RIGHT click opens a menu of choices. You need it in one minute. Practise: right-click on the desktop. A menu should appear. Press Escape to close it.

Teacher: Do not skip this. The next step needs a right-click, and a student who cannot right-click is stuck for the whole period. Watch hands, not screens. The ones who struggle are using one finger on a whole trackpad. These are brand new machines. Say it now: we treat them with respect.

04-10 min | Best performance

Students: RIGHT-click the battery icon at the bottom right. Choose Power and battery settings. Find the power mode boxes. There are TWO: On battery, and Plugged in. Set BOTH of them to Best performance. Stay on this screen. You need it next.

Teacher: Both boxes. Students set the plugged-in one and stop, then wonder why it is slow on the cart. Right-clicking the battery icon is faster than digging through Settings. That is why the trackpad step came first. Keep the window open, and maximize it, for the night light step.

10-17 min | Night light

Students: Maximize this window so you can see the whole thing. At the top there is a search box that says Find a setting. Type: night light. Toggle Night light ON. Slide it ALL the way to the right. Look at how orange that is. Now bring it back to 25 percent. Close Settings.

Teacher: Explain WHY while they are looking at full orange: screens put out blue light, blue light is the part that tires your eyes and gives some people headaches, and night light takes the blue out. Full orange first is what makes 25 percent stick. They need to see the extreme. Walk the room and check screens are actually near 25, not left on full or switched back off.

17-23 min | Teams settings

Students: Open Microsoft Teams and sign in with your school account. Click the three dots next to your profile picture, then Settings. You will see FOUR checkboxes. Turn them all OFF. They are about auto-starting Teams, running in the background, and staying open when you close it. Read the exact wording off your own screen and check your neighbour.

Teacher: There are four checkboxes. The exact wording shifts between Teams versions, so read the current ones aloud from the projector rather than trusting a printed list. The point is that Teams should not launch itself or keep running after it is closed, on a shared cart machine. Expect a slow first sign-in. Clicking repeatedly makes it slower.

23-31 min | The blue cloud

Students: Open OneDrive and look at the cloud icon. BLUE cloud means you are signed in. You are done, help someone near you. GREY cloud with a line through it means you are NOT signed in. If it is grey: LEFT-click it once and follow each prompt. Next, Next, Use this folder, Keep yes. Hand up if it will not let you in.

Teacher: Blue means in, grey with a line means out. Give them that one check and they can diagnose it themselves all year. This is the step that splits the class. Send the blue-cloud students to sit beside the grey-cloud ones. Nobody signs in as anybody else. A student who cannot get in pairs up and watches.

31-33 min | Check each other

Students: Check your partner: night light at 25, both power boxes on Best performance, blue cloud. Only if there is time left, add the printer your teacher wrote on the board.

Teacher: The printer is the lowest priority. Drop it rather than rushing the OneDrive step. Collect who still has a grey cloud. That is your follow-up list.

33-35 min | Respect the machine

Students: These laptops are brand new. Close the lid gently with two hands. Back on the cart, plugged in, cable not crushed. Tell your teacher the one step that did not work for you.

Teacher: Say it plainly: these are new, they are shared, and they are not replaced if they are broken. Write down what each step actually took. The times in this lesson are estimates until you run it once.

What to collect or hear

Show the teacher your blue OneDrive cloud and Teams open.

Success check

These laptops are brand new. Close the lid gently with two hands. Back on the cart, plugged in, cable not crushed. Tell your teacher the one step that did not work for you.

Support and stretch

Grade 5: Follow along step by step with the projected screen.

Grade 6: Help a neighbor after your blue cloud check.

Another way to do it

Pair with a neighbor and watch. Nobody signs in for anyone else.

Optional next challenge

Help someone at your table who is not finished yet, then check your own night light is back at 25 percent.

Cleanup

Check your partner: night light at 25, both power boxes on Best performance, blue cloud.

Only if there is time left, add the printer your teacher wrote on the board.

Teacher notes

Run this once per class, before any lesson that needs OneDrive or printing. It reliably fills a 35 minute period, so do not plan other content on top of it.

The step times below are ESTIMATES for a first run. Use the class clock to record what each step actually took, then adjust. The point of the first run is measurement.

OneDrive is the step that splits the class. Blue cloud means signed in, grey cloud with a line means not. Give students that check and they can diagnose it themselves all year. Pair blue-cloud students with grey-cloud ones rather than waiting.

Teach left click and right click FIRST. The very next step needs a right-click on the battery icon, so a student who cannot right-click is stuck for the whole period.

The printer is the lowest priority and is now the only optional item. Drop it rather than rushing OneDrive. If you do want it, write the hallway printer name on the board first; grade 5 and grade 6 are different.

L50 | Library launch: find it and ask a question

GOAL: I can use call numbers or shelf signs to find a book and write a question it could answer.

35 minutes. Lesson reference: year:b-library-launch

Before students arrive

Call number cards

Question slips

Checkout ready only if it runs today

Materials

Library shelves and signs; Existing library catalog on a teacher device; Call number cards; Question slips and pencils; Existing checkout system, if checkout runs today

Safety

Walk between shelves. Put books back where you found them or on the return cart.

Run the period

00-03 min | Get ready

Students: Sit with a partner. Read the goal: find it and ask.

Teacher: Show a call number on the screen.

03-05 min | Tell your partner

Students: Tell a partner about a book you liked.

Teacher: Take one or two answers.

05-10 min | How books are found

Students: Watch how a call number points to a shelf.

Teacher: Walk to one shelf and find a book live.

10-20 min | Find it

Students: Pick a topic you are curious about. Use shelf signs or the catalog to find one book. Write its call number.

Teacher: Help pairs read the signs.

20-28 min | Ask a question

Students: Write one question the book could answer. Check out the book if checkout runs today.

Teacher: Run checkout only if it fits today's schedule.

28-32 min | Reset together

Students: Return books you are not keeping. Push in chairs.

Teacher: Check the shelves you used.

32-35 min | Show your result

Students: Read your question to a partner.

Teacher: Collect question slips.

What to collect or hear

One call number and one question the book could answer.

Success check

Read your question to a partner.

Support and stretch

Grade 5: Find one book with a partner using shelf signs.

Grade 6: Find two books on one topic and compare which answers the question better.

Another way to do it

Printed call number cards and shelf signs only.

Optional next challenge

Find a second book that could answer the same question.

Cleanup

Return books you are not keeping.

Push in chairs.

Teacher notes

Library and information literacy. Checkout is conditional on the real schedule.

No-tech path: use printed call number cards and shelf signs only.

L51 | Fact card 1: Find and check a source

GOAL: I can find three facts in a source I checked and write them in my own words.

35 minutes. Lesson reference: year:b-factcard-research

Before students arrive

Fact sheets with three boxes and a source line

Source check card on the board

Teacher-listed sites or library books ready

Materials

Library books or teacher-chosen websites; Source check card: who made it, when, can I check it; Fact sheet with three boxes and a source line; Pencils

Safety

Use only teacher-listed sites. No sign-ups or downloads.

Prepare and test this first

Preparation required: choose and check the listed sites or books, and make one strong and one weak example source for the Check a source step. This packet does not supply them.

Run the period

00-03 min | Get ready

Students: Get your fact sheet. Read the goal: three checked facts.

Teacher: Show the source check card.

03-04 min | Tell your partner

Students: Tell a partner your topic.

Teacher: Help anyone stuck pick a topic.

04-09 min | Check a source

Students: Look at two sources the teacher shows. Which one would you trust? Why?

Teacher: Compare a strong source and a weak one.

09-23 min | Find three facts

Students: Find three facts in one checked source. Write each fact in your own words.

Teacher: Ask students to read a fact aloud without looking.

23-28 min | Write the source

Students: Write the title, the author or site, and the date.

Teacher: Check source lines on three sheets.

28-32 min | Save your notes

Students: Put your fact sheet in the class folder. Return books and devices.

Teacher: Collect sheets for next time.

32-35 min | Show your result

Students: Point to one fact you wrote in your own words.

Teacher: Accept pointing or reading aloud.

What to collect or hear

Your fact sheet with three facts in your own words and the source.

Success check

Point to one fact you wrote in your own words.

Support and stretch

Grade 5: Use one book or one listed site. Copy the title and author.

Grade 6: Compare two sources and explain which is more trustworthy.

Another way to do it

Use library books only, same fact sheet.

Optional next challenge

Find a fourth fact and check it in a second source.

Cleanup

Put your fact sheet in the class folder.

Return books and devices.

Teacher notes

Library and digital literacy. Students choose a topic they are into, then check the source before taking facts.

No-tech path: books only, same fact sheet.

L52 | Fact card 2: Design it and share

GOAL: I can design a clear fact card with my three facts and source, then share it with my teacher.

35 minutes. Lesson reference: year:b-factcard-design

Before students arrive

Laptop cart charged and counted

Fact sheets from last time

Paper and markers for the fallback

Materials

Existing school devices on the cart; School Canva through resources.svsd.net; Fact sheets from last time; Paper and markers for the fallback

Safety

Use school Canva only. No new accounts, purchases or downloads. Share only with the teacher.

Run the period

00-04 min | Sign in

Students: Get your device. Open resources.svsd.net, then Canva. Sign in with your school account.

Teacher: Help one table at a time. Switch sign-in problems to paper.

04-07 min | Read a good card

Students: Look at a fact card shown small. What makes it easy to read?

Teacher: Listen for big title, few colors, short lines.

07-21 min | Design your card

Students: Start a blank design or a simple template. Add your three facts and your source line. Use two colors and one clear font.

Teacher: No purchases. Skip anything marked with a crown.

21-25 min | Partner check

Students: Trade screens. Can your partner read it from a step back? Fix one thing.

Teacher: Ask what they fixed.

25-29 min | Share with teacher

Students: Choose Share. Add your CIRC teacher's school account. It is turned in when your teacher can open it.

Teacher: Keep public and link sharing unchanged. Check that you can open it from your school account.

29-32 min | Devices back

Students: Close Canva. Put your device back on the cart and plug it in.

Teacher: Count the cart.

32-35 min | Show your result

Students: Tell a partner which fact is most surprising.

Teacher: Paper cards go in the class tray.

What to collect or hear

Share with your CIRC teacher's school account. It is turned in when your teacher can open it. Keep public and link sharing unchanged.

Success check

Tell a partner which fact is most surprising.

Support and stretch

Grade 5: Two colors, one font, three facts and the source line.

Grade 6: Explain one design choice that makes the card easier to read.

Another way to do it

Make the card on paper with the same checklist.

Optional next challenge

Add a small picture or icon that matches one fact.

Cleanup

Choose Share.

Add your CIRC teacher's school account.

It is turned in when your teacher can open it.

Close Canva.

Put your device back on the cart and plug it in.

Teacher notes

Digital literacy. Same rule as the Canva logo visits: it is turned in when you can open it from your school account. A name in Share alone is not enough.

If Canva or sign-in fails: make the card on paper with the same checklist.

L53 | Mystery Materials

GOAL: Test unknown material samples and select the best candidate for a stated design need.

35 minutes. Lesson reference: catalog:33

Before students arrive

Coded material samples

Droppers, small loads and data tables

Materials

Coded material samples; Droppers; Small loads; Rulers; Scratch-free test tools; Data table

Safety

Use only teacher-prepared samples and approved tests. Do not taste, burn, tear open, or identify by smell.

Run the period

00-03 min | Get ready

Students: Place coded samples on the test mat. Open the data table.

Teacher: A material is useful because of its properties in context. One test rarely proves everything, so choose tests connected to the actual need.

03-06 min | Sample safety

Students: Use only approved tests. Keep dropper water inside the tray. Wipe each spill with a towel. Taste or smell no sample.

Teacher: Model labeling samples without naming their identities.

06-11 min | Choose success rules

Students: Read the design need. Choose two things you can measure. Write the success levels.

Teacher: Demonstrate safe flexibility, absorbency, and load tests. Provide criteria that require more than one property.

11-18 min | Run test one

Students: Test one coded sample at a time. Measure the result. Record every code.

18-25 min | Run test two

Students: Use the second approved property test. Measure each sample again. Record every result.

25-30 min | Choose a material

Students: Compare all results with the success rules. Choose one coded sample. Reject one alternative with evidence.

30-32 min | Reset the samples

Students: Return every coded sample. Dry the test tray.

32-35 min | Exit check

Students: Write the chosen code. Cite two measured results.

What to collect or hear

Your chosen material with two test results and one tradeoff.

Success check

Submit the chosen material with two test results and one acknowledged tradeoff.

Support and stretch

Grade 5: Test two properties and choose one material.

Grade 6: Explain one tradeoff of your choice.

Another way to do it

Compare samples by touch, bend and weight only.

Optional next challenge

Composite Idea: Combine two sample types on paper and explain how their properties could complement each other.

Cleanup

Dry samples, return them by code, and clean test tools without revealing identities to other teams.

Teacher notes

A material is useful because of its properties in context.

One test rarely proves everything, so choose tests connected to the actual need.

Model labeling samples without naming their identities.

Demonstrate safe flexibility, absorbency, and load tests.

Provide criteria that require more than one property.

L54 | Hull Design

GOAL: Shape a foil hull that stays stable while carrying the greatest tested payload.

35 minutes. Lesson reference: catalog:11

Before students arrive

Water tubs, towels and foil sheets

Cargo pieces counted. Laptops away.

Materials

Aluminum foil sheets; Water tubs; Uniform cargo pieces; Towels; Recording sheet

Safety

Keep water inside tubs, wipe spills immediately, and keep electronics out of the wet zone.

Run the period

00-03 min | Get ready

Students: Place one foil sheet beside the tub. Count the cargo pieces.

Teacher: The hull must push aside enough water to support itself and its cargo. Capacity is useful only when the boat stays balanced.

03-06 min | Water safety

Students: Keep water inside the tub. Wipe each spill with a towel. Keep electronics outside the wet zone.

Teacher: Compare a flat foil sheet with a sealed basin shape.

06-10 min | Plan the hull

Students: Sketch a wide sealed hull. Mark one likely weak point.

Teacher: Model adding identical cargo pieces one at a time.

10-18 min | Shape the foil

Students: Fold the edges above the water line. Press the bottom flat. Use no extra foil.

Teacher: Set towel stations and a clear wet-material boundary.

18-26 min | Float and load

Students: Place the hull on the water. Add one cargo piece at a time. Record the last safe total.

26-31 min | Reshape the hull

Students: Change one hull feature. Repeat the load test.

31-33 min | Dry the station

Students: Remove every cargo piece. Dry the table.

33-35 min | Exit check

Students: Write the better cargo total.

What to collect or hear

Your best cargo count and how hull shape helped.

Success check

Record the best cargo count and explain how hull shape affected stability.

Support and stretch

Grade 5: Count cargo with tally marks.

Grade 6: Compare two hull shapes and explain stability.

Another way to do it

Dry version: fold hulls and predict while the teacher floats one.

Optional next challenge

Cargo Deck: Create a removable paper loading guide that helps another team balance cargo evenly.

Cleanup

Dry cargo pieces and tables, empty tubs only at the assigned sink, and recycle clean foil when possible.

Teacher notes

The hull must push aside enough water to support itself and its cargo.

Capacity is useful only when the boat stays balanced.

Compare a flat foil sheet with a sealed basin shape.

Model adding identical cargo pieces one at a time.

Set towel stations and a clear wet-material boundary.

L55 | Seed Travelers

GOAL: Investigate seed-dispersal strategies and build a model seed that travels by wind, drop, float, or hitchhiking simulation.

35 minutes. Lesson reference: catalog:7

Before students arrive

Seed samples or image cards

Teacher-run fan and a water tray

Materials

Teacher-provided seed samples or image cards; Paper; Coffee filters; Cotton; Yarn; Paper clips; Tape; Fan; Water tray; Fabric test strip

Safety

Use only teacher-provided samples, do not taste any material, keep water at the wet station, and release models only into the marked test area.

Run the period

00-03 min | Get ready

Students: Choose one seed sample card. Place the test sheet on your table.

Teacher: Plants cannot walk, but their seeds use shape, air, water, animals, and gravity to move. A model is successful when its structure matches the travel strategy being tested.

03-06 min | Test safely

Students: Keep water inside the tray. Wipe each spill with a towel. Keep every sample away from your mouth.

Teacher: Show teacher-provided seed samples or image cards without collecting from living plants. Set wind, drop, float-tray, and fabric hitchhiker test stations with clear boundaries.

06-10 min | Sort the seeds

Students: Study the seed shape. Choose wind, drop, float, or hitchhiking. Record your reason.

Teacher: Model matching one seed feature to one travel strategy.

10-19 min | Build a model

Students: Sketch one model seed. Choose materials that match the travel method. Build the model.

19-26 min | Run three trials

Students: Use the matching test station. Run one trial at a time. Record each result.

26-30 min | Change one part

Students: Change one shape or material. Repeat the same test.

30-32 min | Reset the station

Students: Dry the wet station. Return every model piece.

32-35 min | Exit check

Students: Circle the better result. Name the useful feature.

What to collect or hear

Your model seed and its trial results.

Success check

Name the travel strategy, show the trial evidence, and explain how one model feature supported movement.

Support and stretch

Support: Copy one real seed's travel trick.

Stretch: Change one feature and compare two trials.

Another way to do it

Use image cards and drop tests by hand, with no fan or water.

Optional next challenge

New Habitat: Draw the habitat where the model seed would have the best chance to travel and begin growing.

Cleanup

Dry the float station, return samples and image cards, collect model pieces, and sort reusable materials.

Teacher notes

Plants cannot walk, but their seeds use shape, air, water, animals, and gravity to move.

A model is successful when its structure matches the travel strategy being tested.

Show teacher-provided seed samples or image cards without collecting from living plants.

Model matching one seed feature to one travel strategy.

Set wind, drop, float-tray, and fabric hitchhiker test stations with clear boundaries.

L56 | Paper Circuit Signal

GOAL: Build a safe paper circuit that uses one LED to communicate a clear signal or message.

35 minutes. Lesson reference: catalog:9

Before students arrive

Copper tape, LEDs and switched battery holders counted

The teacher checks each circuit before power

Materials

Cardstock; Copper tape; LEDs; Switched 2xAA battery holders; AA batteries; Clear tape; Pencils; Circuit planning cards

Safety

Never connect the battery leads directly. Switch off and report any warm holder immediately. Keep batteries inside the holder and return every holder by count.

Run the period

00-03 min | Get ready

Students: Place the circuit card on your table. Keep the power switch off.

Teacher: A circuit works only when electricity has a complete path from one battery side and back to the other. The light is not decoration today. It must communicate something to a user.

03-07 min | Power safety

Students: Use only the switched battery holder. Keep each battery inside the holder. Switch off any warm holder. Report the warm holder to the teacher.

Teacher: Demonstrate LED polarity, a copper-tape path, and the switched 2xAA battery holder. Show the safe response to a warm battery, short circuit, torn tape, or unlit LED.

07-12 min | Plan the signal

Students: Choose one clear signal word. Draw one complete circuit path. Mark which way the long light leg faces.

Teacher: Approve the paper plan before distributing battery holders.

12-14 min | Teacher check

Students: Hold up your paper plan. Wait until the teacher says yes. Do not take a battery box yet.

14-21 min | Build the circuit

Students: Press copper tape along the path. Make each corner smooth. Place the LED in the marked direction.

21-26 min | Power the signal

Students: Connect the holder leads to the marked points. Turn on the switch. Check the first place where the path breaks.

26-30 min | Finish the message

Students: Test the light three times. Add the signal word after success.

30-32 min | Power down

Students: Turn off the switch. Return the holder by count.

32-35 min | Exit check

Students: Point to the complete circuit path. Name one repair you made.

What to collect or hear

Your signal shown three times and the path traced.

Success check

Demonstrate the signal three times and identify the exact path electricity follows.

Support and stretch

Support: Build the circuit from the planning card.

Stretch: Add a switch and a clear message.

Another way to do it

Draw the circuit path and trace it with a finger while the teacher demonstrates.

Optional next challenge

Switch Upgrade: Design a paper press switch that turns the signal on only when a user intentionally activates it.

Cleanup

Switch off and count battery holders first, sort working LEDs, press loose copper-tape ends flat, and store finished circuits without batteries connected.

Teacher notes

A circuit works only when electricity has a complete path from one battery side and back to the other.

The light is not decoration today. It must communicate something to a user.

Demonstrate LED polarity, a copper-tape path, and the switched 2xAA battery holder.

Show the safe response to a warm battery, short circuit, torn tape, or unlit LED.

Approve the paper plan before distributing battery holders.

L57 | Earthquake Platform

GOAL: Build and improve a model structure that remains stable during a repeatable shake-table test.

35 minutes. Lesson reference: catalog:30

Before students arrive

Shake platforms, or a table shaken by hand with a fixed count

Damage rubric out

Materials

Shake platforms; Craft sticks; Straws; Cards; Tape; Binder clips; Damage rubric

Safety

Keep fingers outside the moving platform boundary and test only on the teacher signal.

Run the period

00-03 min | Get ready

Students: Place the platform on its boundary mat. Check the height rule.

Teacher: Earthquake design cannot stop the ground from moving, but it can control how a structure responds. A fair comparison uses the same platform motion and structure limits.

03-06 min | Platform safety

Students: Keep fingers outside the moving platform. Test only on the teacher signal. Stop any falling part.

Teacher: Model a low center of mass, cross bracing, and flexible connection.

06-10 min | Plan for shaking

Students: Sketch the structure. Mark two likely weak joints. Check the required footprint.

Teacher: Demonstrate the approved shake sequence and damage rubric.

10-19 min | Build the structure

Students: Build the base inside the footprint. Reach the required height. Check every connection.

Teacher: Inspect structures for loose pieces before testing.

19-26 min | Run the shake

Students: Move hands behind the boundary. Run the standard shake sequence. Record movement or damage.

26-30 min | Strengthen one part

Students: Improve one weak connection. Repeat the same shake sequence.

30-32 min | Reset the platform

Students: Remove loose pieces from the platform. Sort reusable parts.

32-35 min | Exit check

Students: Compare both damage scores. Name the stronger feature.

What to collect or hear

Two damage scores and the change that helped.

Success check

Compare the two damage scores and name the structural change supported by evidence.

Support and stretch

Grade 5: Build, shake, fix one weak spot.

Grade 6: Compare damage scores and name the structural change.

Another way to do it

Shake the table by hand with a fixed count.

Optional next challenge

Aftershock: Test whether the improved structure survives a second shake without repairs.

Cleanup

Remove loose pieces from the platform, release clips carefully, and return shake tables to neutral.

Teacher notes

Earthquake design cannot stop the ground from moving, but it can control how a structure responds.

A fair comparison uses the same platform motion and structure limits.

Model a low center of mass, cross bracing, and flexible connection.

Demonstrate the approved shake sequence and damage rubric.

Inspect structures for loose pieces before testing.

L58 | Maze Coders

GOAL: Write an efficient algorithm that moves a token through a grid maze without entering blocked cells.

35 minutes. Lesson reference: catalog:15

Before students arrive

Grid maze cards, tokens and command cards

Materials

Grid maze cards; Tokens; Command cards; Pencils; Transparent sleeves

Safety

Keep all work on the grid surface and keep small tokens away from mouths and floor traffic.

Run the period

00-03 min | Get ready

Students: Place one maze card in the sleeve. Put the token on start.

Teacher: A good route is accurate first and efficient second. Debug from the first wrong move instead of rewriting everything blindly. Provide mazes with a verified solution path.

03-06 min | Token safety

Students: Keep the token on the grid. Pick up any dropped token at once.

Teacher: Model reading row and column coordinates before writing commands.

06-11 min | Map the route

Students: Mark the goal. Mark each blocked cell. Trace one possible route.

Teacher: Run one incorrect path and trace back to the first error.

11-18 min | Write the code

Students: Write one command per box. Finish the full route before testing.

18-24 min | Partner test

Students: Trade code with a partner. Run each command exactly. Mark the first error.

24-29 min | Debug the route

Students: Change the first wrong command. Remove one extra command if possible. Run the route again.

29-32 min | Reset the maze

Students: Return the token. Erase the sleeve.

32-35 min | Exit check

Students: Write the number of final commands. Circle the command you changed.

What to collect or hear

Your final command count and the command that fixed the first bug.

Success check

Submit the final command count and circle the command that fixed the first bug.

Support and stretch

Support: Solve the small maze first.

Stretch: Find a shorter route and prove it.

Another way to do it

This lesson is already device free.

Optional next challenge

Shortest Safe Route: Find a second valid route and prove which one uses fewer commands.

Cleanup

Count tokens, erase sleeves, and sort maze cards by level.

Teacher notes

A good route is accurate first and efficient second.

Debug from the first wrong move instead of rewriting everything blindly.

Model reading row and column coordinates before writing commands.

Run one incorrect path and trace back to the first error.

Provide mazes with a verified solution path.

L59 | Binary Beacons

GOAL: Encode and decode a message using a documented two-state binary system.

35 minutes. Lesson reference: catalog:16

Before students arrive

Two-color cards and binary mats

Message slips

Materials

Two-color cards; Paper cups; Flashlights used at tables; Binary place-value mats; Message slips

Safety

Do not shine lights toward faces and use only classroom-safe, nonpersonal messages.

Run the period

00-03 min | Get ready

Students: Choose two state cards. Place the beacon at table level.

Teacher: Digital systems build complex information from simple states such as on and off. A code is useful only when sender and receiver share the same key.

03-06 min | Light safety

Students: Keep every light pointed at the table. Turn off any light aimed near a face. Use only classroom-safe messages.

Teacher: Model representing small numbers with place-value beacon cards.

06-11 min | Build the key

Students: Assign a value to each state. Write the value in each number spot. Test one sample number.

Teacher: Demonstrate how one missing bit changes a decoded result.

11-18 min | Encode a message

Students: Choose a short classroom message. Convert each symbol with the key. Place the states in order.

Teacher: Require teams to exchange a key before transmitting.

18-24 min | Send the states

Students: Show one fixed position at a time. Pause between symbols. Let the partner record silently.

24-29 min | Check the message

Students: Compare the partner answer with the original message. Correct one unclear state.

29-32 min | Power down

Students: Turn off every light. Stack the state cards.

32-35 min | Exit check

Students: Write one rule that prevented confusion.

What to collect or hear

Your key, coded message and decoded result.

Success check

Provide the key, encoded message, decoded result, and one corrected transmission error.

Support and stretch

Support: Send three letters with a partner.

Stretch: Add an error check and fix one mistake.

Another way to do it

Use cards only, no flashlights.

Optional next challenge

Error Check: Add one check bit or repeated pattern that helps the receiver notice a transmission error.

Cleanup

Turn off lights, stack state cards by color, and destroy temporary message slips if directed.

Teacher notes

Digital systems build complex information from simple states such as on and off.

A code is useful only when sender and receiver share the same key.

Model representing small numbers with place-value beacon cards.

Demonstrate how one missing bit changes a decoded result.

Require teams to exchange a key before transmitting.

L60 | Black Box Systems

GOAL: Use controlled input-output tests to build and defend a model of a hidden system.

35 minutes. Lesson reference: catalog:20

Before students arrive

Sealed systems prepared by the teacher

Input pieces and data tables

Materials

Teacher-prepared sealed systems; Input pieces; Measuring tools; Data table; Modeling paper

Safety

Do not pry, shake forcefully, or place any object into an opening not approved for testing.

Run the period

00-03 min | Get ready

Students: Place the sealed system on its mat. Open the data table.

Teacher: Scientists and engineers often cannot see every part, so they use patterns in evidence. A model is a best explanation that can change when new evidence appears.

03-06 min | Black box safety

Students: Keep the box sealed. Use only the approved test holes. Move the box gently.

Teacher: Demonstrate changing one input while recording the output.

06-10 min | Choose test actions

Students: List three safe test actions. Change only one test action at a time.

Teacher: Model two competing explanations for the same first result.

10-18 min | Run controlled tests

Students: Try the first test action. Measure the result. Record the matching action and result. Repeat each action once.

Teacher: Keep sealed boxes closed so evidence comes only from allowed tests.

18-25 min | Draw the system

Students: Draw one possible inside system. Connect each test action to a result. Label the hidden parts.

25-30 min | Challenge the model

Students: Choose one test that could show the drawing is wrong. Run that test. Revise the drawing if needed.

30-32 min | Reset the box

Students: Remove every test piece. Return the sealed system.

32-35 min | Exit check

Students: Write one claim about the hidden system. Cite one test result.

What to collect or hear

Your model, one supporting result and one question you still have.

Success check

Present the model, one supporting result, and one remaining uncertainty.

Support and stretch

Support: Try three inputs and record outputs.

Stretch: Explain one thing you still cannot tell.

Another way to do it

This lesson is already device free.

Optional next challenge

Competing Model: Draw a different internal system that could produce the same evidence, then design a test that separates the two.

Cleanup

Return each sealed system with its complete input set and submit data separately.

Teacher notes

Scientists and engineers often cannot see every part, so they use patterns in evidence.

A model is a best explanation that can change when new evidence appears.

Demonstrate changing one input while recording the output.

Model two competing explanations for the same first result.

Keep sealed boxes closed so evidence comes only from allowed tests.

L61 | Error-Proof Messages

GOAL: Create a communication system that lets a receiver detect or correct a changed message.

35 minutes. Lesson reference: catalog:21

Before students arrive

Symbol cards, message strips and envelopes

Materials

Symbol cards; Message strips; Envelopes; Check-rule cards; Pencils

Safety

Use no personal, insulting, or secret information in transmitted messages.

Run the period

00-03 min | Get ready

Students: Choose one symbol key. Place the channel envelope between teams.

Teacher: Real messages can be lost, smudged, or changed on the way. Redundancy uses extra information so an error does not stay invisible.

03-06 min | Message safety

Students: Use no personal information. Use no secret information. Use respectful classroom words.

Teacher: Model a short message with and without a simple check rule.

06-11 min | Encode the message

Students: Choose a short safe message. Convert it with the symbol key. Check every symbol once.

Teacher: Change one symbol in transit and compare receiver confidence.

11-18 min | Add an error check

Students: Choose repeat, number-sum check, even-or-odd check, or pattern. Add the check to the message. Write the checking rule.

Teacher: Keep challenge messages classroom-safe and anonymous.

18-24 min | Send through noise

Students: Give the message to the change team. Let the channel change one part. Receive the changed message.

24-29 min | Detect the error

Students: Apply the checking rule. Mark the possible error. Decode the message.

29-32 min | Reset the channel

Students: Return every symbol card. Recycle used message strips.

32-35 min | Exit check

Students: Name the error check. State whether it found the change.

What to collect or hear

The changed message and how the receiver caught it.

Success check

Show the altered message and explain exactly how the receiver detected or corrected it.

Support and stretch

Support: Use one simple check rule.

Stretch: Design a check that fixes the error, not just finds it.

Another way to do it

This lesson is already device free.

Optional next challenge

Repair Mode: Improve the system so the receiver can correct one missing symbol, not only notice it.

Cleanup

Return reusable code cards and recycle or destroy temporary message strips as directed.

Teacher notes

Real messages can be lost, smudged, or changed on the way.

Redundancy uses extra information so an error does not stay invisible.

Model a short message with and without a simple check rule.

Change one symbol in transit and compare receiver confidence.

Keep challenge messages classroom-safe and anonymous.

L62 | Biomimicry Grabber

GOAL: Design a gripping mechanism inspired by how a plant or animal holds, hooks, wraps, or adheres.

35 minutes. Lesson reference: catalog:32

Before students arrive

Six table trays stocked, 30 scissors counted

Nature reference cards

Materials

Nature reference cards; Cardboard; Craft sticks; String; Rubber bands; Foam; Lightweight test objects; Scissors

Safety

Use no live organism, body grip test, sharp hook, or adhesive on skin or personal property.

Run the period

00-03 min | Get ready

Students: Choose one nature reference card. Place lightweight objects on the test mat.

Teacher: Biomimicry copies a useful strategy, not only the way an organism looks. Name the function in nature before turning it into a design feature.

03-06 min | Grabber safety

Students: Cut only on the table. Keep fingers outside the cutting path. Carry scissors closed. Test no body or live organism.

Teacher: Show image examples of hooks, suction, wrapping, and textured grip.

06-11 min | Copy a strategy

Students: Name the nature trick. Name its useful function. Sketch a safe gripper.

Teacher: Model translating one observed function into a simple mechanism.

11-19 min | Build the grabber

Students: Cut only planned cardboard shapes. Build the moving connection. Add a soft place that can hold an object.

Teacher: Approve lightweight test objects and movement ranges.

19-26 min | Test three objects

Students: Grip the first approved object. Record the result. Repeat for two more objects.

26-30 min | Improve the grip

Students: Change one nature-inspired part. Repeat one failed object test.

30-32 min | Reset the grabber

Students: Close the scissors. Return every test object.

32-35 min | Exit check

Students: Name the copied strategy. Cite one test result.

What to collect or hear

Your grabber and the animal or plant idea it copies.

Success check

Demonstrate the grip and connect one design feature to the biological function it models.

Support and stretch

Support: Copy one hook or wrap from a nature card.

Stretch: Test two grips and keep the better one.

Another way to do it

Draw the grabber and explain the nature idea.

Optional next challenge

Habitat Shift: Adapt the gripper for an object with a different surface, shape, or location.

Cleanup

Return reference cards, release stretched bands, and sort test objects into their labeled set.

Teacher notes

Biomimicry copies a useful strategy, not only the way an organism looks.

Name the function in nature before turning it into a design feature.

Show image examples of hooks, suction, wrapping, and textured grip.

Model translating one observed function into a simple mechanism.

Approve lightweight test objects and movement ranges.

L63 | Rube Machine

GOAL: Build a reliable chain reaction in which at least three distinct steps complete one simple task.

35 minutes. Lesson reference: catalog:13

Before students arrive

Dominoes, ramps, cups and balls in trays

Final task cards out

Materials

Dominoes; Cardboard ramps; Cups; String; Craft sticks; Balls; Tape; Final-task cards

Safety

Keep moving parts inside the table boundary and use no flame, sharp projectile, or powered launcher.

Run the period

00-03 min | Get ready

Students: Choose one final task card. Place moving parts inside the table boundary.

Teacher: Complex systems succeed one dependable connection at a time. The goal is not extra chaos. The goal is a clear chain that another team can explain.

03-06 min | Motion safety

Students: Use only rolling tabletop parts. Keep hands outside a moving path. Stop any part that leaves the table.

Teacher: Model two safe cause-and-effect connections.

06-11 min | Plan backward

Students: Draw the final task first. Add three earlier steps. Mark where each part starts the next part.

Teacher: Show how to isolate and test one step before joining the whole machine.

11-19 min | Test each step

Students: Build one step. Test that step one time. Fix the first part that does not work.

Teacher: Set boundaries for rolling objects and moving parts.

19-27 min | Connect the chain

Students: Join the tested steps. Try one full run. Fix the first place where motion stops. Try one more full run if time remains.

27-32 min | Reset the parts

Students: Stop every moving part. Sort parts into labeled bins. Remove tape scraps.

32-35 min | Exit check

Students: Circle the hardest move from one part to the next. Write the fix that worked.

What to collect or hear

Two complete runs and the handoff you fixed.

Success check

Demonstrate two complete runs and explain the handoff that required the most revision.

Support and stretch

Support: Three steps that work once.

Stretch: Two complete runs in a row.

Another way to do it

Plan the chain on paper and act it out.

Optional next challenge

One More Link: Add one useful step without reducing the machine's two-run reliability.

Cleanup

Stop all motion, collect small rolling parts first, and sort components into labeled bins.

Teacher notes

Complex systems succeed one dependable connection at a time.

The goal is not extra chaos. The goal is a clear chain that another team can explain.

Model two safe cause-and-effect connections.

Show how to isolate and test one step before joining the whole machine.

Set boundaries for rolling objects and moving parts.

L64 | Check the AI answer

GOAL: I can check claims in an AI-style answer against a trusted source and fix what is wrong.

35 minutes. Lesson reference: year:b-ai-check

Before students arrive

Printed AI answer cards prepared by the teacher

Library books or teacher-listed sites

Highlighters

Materials

Printed AI answer cards the teacher prepared; Library books or teacher-listed sites; Highlighters or colored pencils; Pencils

Safety

No student AI accounts. Students read printed answers only.

Prepare and test this first

Preparation required: write the AI answer cards with a mix of right and wrong claims, check each claim in a named book or listed site, and keep an answer key. This packet does not supply card text, sources or a key.

Run the period

00-03 min | Get ready

Students: Get your AI answer card. Read the goal: check, then fix.

Teacher: Hand out cards and highlighters.

03-04 min | Tell your partner

Students: Tell a partner a time something online was wrong.

Teacher: Take one answer.

04-09 min | Watch a check

Students: AI tools can sound sure and still be wrong. Watch the teacher check one claim.

Teacher: Check one claim live in a book or listed site.

09-22 min | Check the claims

Students: Read your AI answer card. Find each claim in a book or checked source. Mark it: matches, wrong, or cannot find.

Teacher: Ask: where did you find that?

22-28 min | Fix the answer

Students: Rewrite one wrong claim correctly. Add where you found it.

Teacher: Look for a source on every fix.

28-32 min | Reset together

Students: Return books. Put your card in the class folder.

Teacher: Collect cards.

32-35 min | Show your result

Students: Read your fixed claim and its source.

Teacher: Accept reading aloud or pointing.

What to collect or hear

Your checked card with one claim fixed and its source.

Success check

Read your fixed claim and its source.

Support and stretch

Grade 5: Check two claims with a partner.

Grade 6: Check every claim and explain why the wrong one might sound right.

Another way to do it

The whole lesson runs on paper with books.

Optional next challenge

Write one question you would ask an AI tool so the answer is easier to check.

Cleanup

Return books.

Put your card in the class folder.

Teacher notes

Library and information literacy, including checking AI answers. Prepare cards ahead with a mix of right and wrong claims.

No-tech path: the whole lesson runs on paper with books.

L65 | Stormwater Rescue

GOAL: Model stormwater runoff and erosion, then test one intervention that slows, redirects, absorbs, or filters the flow.

35 minutes. Lesson reference: catalog:27

Before students arrive

Sloped trays, measured water cups and towels

Wet zone taped. Laptops away.

Materials

Sloped trays; Model soil or sand; Measured water cups; Collection cups; Gravel; Sponge; Craft sticks; Mesh; Model plants; Towels; Scissors; Wet mats; Return tub

Safety

Never drink test water, keep wet materials inside trays, wipe spills immediately, and wash hands after handling model soil.

Run the period

00-03 min | Get ready

Students: Place the sloped tray on the wet mat. Measure the rain water.

Teacher: Rain becomes a design problem when water moves too quickly across a surface and carries soil with it. The goal is not drinkable water. The goal is evidence that one intervention changes runoff or erosion in the model.

03-07 min | Stormwater safety

Students: Keep wet material inside the tray. Wipe each spill with a towel. Cut mesh only on the table. Keep fingers outside the cutting path. Taste no test material.

Teacher: Set identical sloped trays and model the baseline rain test with a fixed water volume and pour time.

07-12 min | Run the first test

Students: Pour the measured rain from the marked height. Measure the water that leaves the tray. Mark where soil moved.

Teacher: Show how to measure runoff volume, visible soil movement, or time to collection.

12-18 min | Build a rescue

Students: Choose one problem location. Cut only the planned mesh piece. Build one helper that slows or moves the water.

Teacher: Demonstrate one barrier, absorbent zone, channel, or plant-root model without presenting a single correct answer.

18-24 min | Repeat the rain

Students: Pour the same water amount. Measure the water that leaves the tray. Mark where soil moved this time.

24-27 min | Improve the rescue

Students: Change one part of the helper. Predict the next result.

27-32 min | Clean the tray

Students: One table at a time, carry wet soil to the return tub. Place wet tools on the drying mat. Dry the tray and table. Wash hands when the teacher calls your table.

32-35 min | Exit check

Students: Compare both water amounts. Name the helpful feature.

What to collect or hear

Baseline and revised results, and what your change did.

Success check

Compare baseline and revised evidence, then state what the intervention changed and what the model cannot prove.

Support and stretch

Grade 5: Compare before and after with a measuring cup.

Grade 6: Explain what the model cannot prove.

Another way to do it

Watch one teacher demo and predict with drawings.

Optional next challenge

Storm Upgrade: Predict how the intervention would perform with twice the water and identify the first likely failure.

Cleanup

Pour water only into the assigned container, return reusable media, dispose of model soil as directed, and dry the full wet zone.

Teacher notes

Rain becomes a design problem when water moves too quickly across a surface and carries soil with it.

The goal is not drinkable water. The goal is evidence that one intervention changes runoff or erosion in the model.

Set identical sloped trays and model the baseline rain test with a fixed water volume and pour time.

Show how to measure runoff volume, visible soil movement, or time to collection.

Demonstrate one barrier, absorbent zone, channel, or plant-root model without presenting a single correct answer.

L66 | Code 1: Draw with a sequence

GOAL: I can put code blocks in order to show pictures in the order I planned.

35 minutes. Lesson reference: year:b-code-sequence

Before students arrive

Laptop cart charged and counted

MakeCode simulator link on the board, if the site is allowed

Materials

Cart laptops, charged; MakeCode for micro:bit in the browser, simulator only, if approved; 5 by 5 grid cards and pencils for the no-device path

Safety

Use only the MakeCode link the teacher gives. No sign-in is needed for the simulator.

Run the period

00-04 min | Open MakeCode

Students: Get your laptop. Open the MakeCode link on the board.

Teacher: Show the link big on the screen.

04-05 min | Tell your partner

Students: Tell a partner a set of steps you follow every day.

Teacher: Link steps to order.

05-09 min | Watch the blocks

Students: Watch blocks snap together in order. Watch the simulator run them.

Teacher: Swap two blocks to show order matters.

09-23 min | Draw with code

Students: Use show leds to make a picture. Add a second picture after it. Run it in the simulator.

Teacher: Ask students to say the order out loud.

23-27 min | Check the order

Students: Does it show in the order you planned? Swap two blocks and watch what changes.

Teacher: Point out one good bug fix.

27-32 min | Save and shut

Students: Name your project with your initials. Close the lid with two hands. Put the laptop back and plug it in.

Teacher: Count the cart.

32-35 min | Show your result

Students: Tell a partner what your pictures show, in order.

Teacher: Accept a spoken answer.

What to collect or hear

A program that shows your pictures in the order you planned.

Success check

Tell a partner what your pictures show, in order.

Support and stretch

Grade 5: Two pictures in order.

Grade 6: Three pictures with a pause, and predict the result before running.

Another way to do it

Write the steps on 5 by 5 grid cards and have a partner run them.

Optional next challenge

Add a third picture and a pause between each.

Cleanup

Name your project with your initials.

Close the lid with two hands.

Put the laptop back and plug it in.

Teacher notes

Computer science. The simulator needs no hardware. Confirm the site is allowed on student laptops before this block.

No-device path: students write the steps on 5 by 5 grid cards and a partner runs them.

L67 | Code 2: Loops make it move

GOAL: I can use a loop to repeat pictures and make an animation.

35 minutes. Lesson reference: year:b-code-loops

Before students arrive

Laptop cart charged and counted

MakeCode simulator link on the board

Materials

Cart laptops, charged; MakeCode for micro:bit in the browser, simulator only, if approved; Grid cards for the no-device path

Safety

Use only the MakeCode link the teacher gives.

Run the period

00-04 min | Open MakeCode

Students: Get your laptop. Open the MakeCode link.

Teacher: Show the link big on the screen.

04-05 min | Recap

Students: Tell a partner why order matters.

Teacher: Listen for one clear answer.

05-09 min | Watch a loop

Students: Watch forever repeat the same steps.

Teacher: Show forever, then repeat 3 times.

09-23 min | Make it move

Students: Make three pictures that change a little. Put them in a forever loop. Add a pause between them.

Teacher: Ask: what repeats, and what changes?

23-27 min | Debug

Students: Find one thing that looks wrong. Fix it and run again. Show your animation to a partner.

Teacher: Celebrate one found bug.

27-32 min | Save and shut

Students: Name your project with your initials. Close the lid with two hands. Put the laptop back and plug it in.

Teacher: Count the cart.

32-35 min | Tell what repeats

Students: Tell a partner what repeats in your code.

Teacher: Accept a spoken answer. Laptops are already put away.

What to collect or hear

An animation that repeats in a loop.

Success check

Tell a partner what repeats in your code.

Support and stretch

Grade 5: Three pictures in a forever loop.

Grade 6: Use a repeat loop with a count and explain how it differs from forever.

Another way to do it

Flip-book grid cards run by a partner three times.

Optional next challenge

Change the pause to speed up or slow down the animation.

Cleanup

Name your project with your initials.

Close the lid with two hands.

Put the laptop back and plug it in.

Teacher notes

Computer science. Loops and debugging.

No-device path: flip-book grid cards run by a partner, repeated three times.

L68 | Code 3: If this, then that

GOAL: I can make a program react differently when different buttons are pressed.

35 minutes. Lesson reference: year:b-code-conditions

Before students arrive

Laptop cart charged and counted

MakeCode simulator link on the board

Materials

Cart laptops, charged; MakeCode for micro:bit in the browser, simulator only, if approved; Paper if-then cards for the no-device path

Safety

Use only the MakeCode link the teacher gives.

Run the period

00-04 min | Open MakeCode

Students: Get your laptop. Open the MakeCode link.

Teacher: Show the link big on the screen.

04-05 min | Recap

Students: Tell a partner what a loop does.

Teacher: Listen for repeat.

05-09 min | Watch a button

Students: Watch on button A pressed change the picture.

Teacher: Click A and B in the simulator.

09-23 min | Make it react

Students: Show one picture when A is pressed. Show another when B is pressed. Test both buttons in the simulator.

Teacher: Ask: what should happen if nobody presses?

23-27 min | Trade and test

Students: Test a partner's program. Tell them one bug or one win.

Teacher: Model kind, specific feedback.

27-32 min | Save and shut

Students: Name your project with your initials. Close the lid with two hands. Put the laptop back and plug it in.

Teacher: Count the cart.

32-35 min | Show your result

Students: Say what happens when A is pressed.

Teacher: Accept a spoken answer.

What to collect or hear

A program that reacts differently to button A and button B.

Success check

Say what happens when A is pressed.

Support and stretch

Grade 5: Button A and button B show different pictures.

Grade 6: Add an if and else condition and test every path.

Another way to do it

Play if-then card games that follow the same rules.

Optional next challenge

Add a shake event with its own picture.

Cleanup

Name your project with your initials.

Close the lid with two hands.

Put the laptop back and plug it in.

Teacher notes

Computer science. Events and conditions, then testing a partner's program.

No-device path: partners play if-then card games that follow the same rules.

L69 | Micro:bit Sensor Station

GOAL: Program a classroom micro:bit or confirmed equivalent to detect a condition and produce a useful visual, sound, or motion alert.

35 minutes. Lesson reference: catalog:28

Before students arrive

Micro:bits or the MakeCode simulator, if available

Paper code blocks for the fallback

Materials

Classroom micro:bits or confirmed equivalents; USB cables; Computers; Teacher-approved sensors if available; Starter code cards; Sensor data cards; Paper code blocks; Test record

Safety

Use only approved classroom hardware and USB power, connect no unknown devices, collect no names or personal sound recordings, and handle sensors by the demonstrated edges.

Run the period

00-05 min | Get ready

Students: Place the micro:bit beside the computer. Open the teacher starter file. Connect only the approved cable. Wait until the computer shows the board.

Teacher: A sensor turns a condition in the world into data a program can use. A useful alert has a clear threshold, a clear user, and a response that can be tested. Use device-free sensor data cards and paper code blocks as the fallback if hardware or network access is unavailable.

05-09 min | Device safety

Students: Use only approved USB power. Connect no unknown device. Handle the board by its edges. Disconnect power when the teacher says stop.

Teacher: Confirm the available micro:bit or equivalent hardware and demonstrate one input, threshold rule, and output.

09-14 min | Plan the sensor

Students: Choose one classroom condition. Write what the board will measure. Choose the number that turns the alert on. Choose a clear alert.

Teacher: Provide starter code blocks for light, temperature, sound, movement, or a teacher-approved external sensor.

14-20 min | Build the program

Students: Place the measure block first. Add the number check. Add the alert block. Transfer the program to the approved device.

20-26 min | Test three values

Students: Test one number below the turn-on number. Test the turn-on number. Test one number above it. Record each alert.

26-29 min | Tune the alert

Students: Change the turn-on number or alert. Repeat the three tests.

29-32 min | Disconnect safely

Students: Stop the program. Unplug the USB cable by its connector. Return the board and cable to the tray.

32-35 min | Exit check

Students: State the final turn-on number. Name one limit of the data.

What to collect or hear

Three input tests and your sensor, threshold and alert.

Success check

Demonstrate three input tests and explain the sensor, threshold, output, and one revision.

Support and stretch

Grade 5: Show an alert when a button or sensor changes.

Grade 6: Choose a threshold and explain why.

Another way to do it

Use paper code blocks to plan and run the program by hand.

Optional next challenge

Two-Level Alert: Add one warning level before the main threshold so the user can respond earlier.

Cleanup

Stop programs, disconnect USB cables by the plug, return every device by number, and leave starter files without student names.

Teacher notes

A sensor turns a condition in the world into data a program can use.

A useful alert has a clear threshold, a clear user, and a response that can be tested.

Confirm the available micro:bit or equivalent hardware and demonstrate one input, threshold rule, and output.

Provide starter code blocks for light, temperature, sound, movement, or a teacher-approved external sensor.

Use device-free sensor data cards and paper code blocks as the fallback if hardware or network access is unavailable.

L70 | Cardboard Arcade

GOAL: Build a small tabletop cardboard skill game with one clear goal, a reliable mechanism, and a fast reset.

35 minutes. Lesson reference: catalog:10

Before students arrive

Six table trays stocked, 30 scissors counted

Contained game tokens counted

Materials

Cardboard; Paper cups; Craft sticks; Paper fasteners; String; Tape; Markers; Marbles or contained game tokens; Score cards; Tabletop game cards; Play trays

Safety

Keep every moving token inside the tabletop play area, use no launchers, and stop any mechanism that pinches or sends parts toward people.

Run the period

00-03 min | Get ready

Students: Choose one tabletop game card. Place tokens inside the play tray.

Teacher: A fun game is a system: the player understands the goal, the mechanism behaves predictably, and the next player can reset it. Start with one playable action, not a giant theme.

03-06 min | Game safety

Students: Keep tokens inside the table boundary. Stop any part that pinches. Use no launcher.

Teacher: Show compact examples such as a marble maze, spinner target, ring challenge, ramp, or contained token drop.

06-10 min | Plan the game

Students: Write the game goal in one sentence. Sketch the token path.

Teacher: Model a start state, player action, scoring rule, and ten-second reset.

10-19 min | Build the game

Students: Build the smallest playable version. Add one reliable cardboard connection. Make a clear start point.

Teacher: Stop teams after the first playable version for a partner test before decoration.

19-26 min | Play test

Students: Ask a partner to play without help. Watch the first confusing moment. Record one unreliable part.

26-30 min | Improve the reset

Students: Change the confusing part. Reset the game twice.

30-32 min | Reset the arcade

Students: Count every token. Place the game on the storage shelf.

32-35 min | Exit check

Students: State the goal. Show how the game resets.

What to collect or hear

A playable game with a ten-second reset.

Success check

Let a partner play, show the ten-second reset, and name the revision that improved the experience.

Support and stretch

Grade 5: One goal and one mechanism.

Grade 6: Add a scoring rule and a ten-second reset.

Another way to do it

Make a paper board game with the same rules.

Optional next challenge

Level Two: Add one optional rule that makes the same game more challenging without rebuilding it.

Cleanup

Return shared tokens and tools by count, label stored games, recycle loose scraps, and leave every game in its reset position.

Teacher notes

A fun game is a system: the player understands the goal, the mechanism behaves predictably, and the next player can reset it.

Start with one playable action, not a giant theme.

Show compact examples such as a marble maze, spinner target, ring challenge, ramp, or contained token drop.

Model a start state, player action, scoring rule, and ten-second reset.

Stop teams after the first playable version for a partner test before decoration.

L71 | Outdoor Habitat Helper

GOAL: Observe a local habitat need and design a safe, realistic helper for biodiversity, water, shelter, observation, or protective communication.

35 minutes. Lesson reference: catalog:31

Before students arrive

Habitat photo cards and site maps

Approval checklist out

Materials

Clipboards; Local habitat photo cards; Site map; Cardboard; Paper; Craft sticks; String; Tape; Rulers; Approval checklist; Scissors; Approval tray

Safety

Observe without touching wildlife, nests, insects, unknown plants, or droppings. No outdoor installation, food, water, digging, or attachment occurs without teacher approval.

Run the period

00-03 min | Get ready

Students: Clip the site map to your board. Choose one observation zone.

Teacher: Habitat design begins with what is already living there and what should not be disturbed. A helper must support a real need without creating a new risk for plants, animals, or people. Lead a no-touch habitat observation or provide local habitat photo cards when outdoor work is unavailable.

03-05 min | Walk out

Students: Walk with your habitat team.

05-09 min | Habitat safety

Students: Observe wildlife without touching it. Leave nests, plants, droppings, or insects alone. Cut model parts only on the table. Keep fingers outside the cutting path. Install nothing outdoors.

09-15 min | Record the need

Students: Record one habitat observation. Name one need supported by that evidence. Mark the location on the map.

Teacher: Model separating observed evidence from guesses about an organism's needs.

15-17 min | Return inside

Students: Walk back with your habitat team. Place the map at your seat.

17-22 min | Plan a helper

Students: Choose one helper type. List one benefit. List one risk. List one maintenance need.

Teacher: Present approval constraints for pollinator support, runoff control, shelter models, observation stations, or protective signs.

22-28 min | Build a small model

Students: Cut only approved model pieces. Build the helper at table scale. Add one clear safety label.

28-31 min | Pack the model

Students: Close the scissors. Place the model in the approval tray.

31-35 min | Exit check

Students: Point to the evidence on your map. Explain why teacher approval is still needed.

What to collect or hear

Your observation, helper, benefit and risk.

Success check

Show the observation, helper, benefit, risk, and the approval needed before any real-world action.

Support and stretch

Grade 5: Pick one habitat need and build one helper.

Grade 6: Name one risk and the approval needed before any real action.

Another way to do it

Use photo cards indoors.

Optional next challenge

Season Check: Explain how the helper may need to change in winter, spring, summer, or fall.

Cleanup

Return observation tools, remove every model material from outdoors, and store proposals separately from approved actions.

Teacher notes

Habitat design begins with what is already living there and what should not be disturbed.

A helper must support a real need without creating a new risk for plants, animals, or people.

Lead a no-touch habitat observation or provide local habitat photo cards when outdoor work is unavailable.

Model separating observed evidence from guesses about an organism's needs.

Present approval constraints for pollinator support, runoff control, shelter models, observation stations, or protective signs.

L72 | Outdoor Classroom Redesign

GOAL: Survey the existing outdoor classroom and design one realistic improvement for learning, accessibility, organization, shade, seating, or habitat.

35 minutes. Lesson reference: catalog:3

Before students arrive

Clipboards, site maps and tape measures counted

Outdoor boundaries decided, weather checked

Materials

Clipboards; Site maps; Grid paper; Pencils; Rulers; Tape measures; Cardboard; Tape; Markers

Safety

Stay inside outdoor boundaries, use measuring tools only as demonstrated, and do not climb, lift, dig, or disturb plants and animals.

Run the period

00-03 min | Get ready

Students: Clip the site map to your board. Choose one outdoor zone.

Teacher: Good redesign begins with what the space already does well and what gets in the way. A useful idea names the user, the need, the location, and how it could actually work.

03-05 min | Walk out

Students: Walk with your design team. Stop at the assigned zone.

05-08 min | Measure safely

Students: Keep the tape measure below your waist. Leave plants untouched. Stay on the approved surface.

Teacher: Revisit the outdoor boundaries and model observing a space without disturbing it.

08-16 min | Survey the zone

Students: Record one feature that works. Record one learning problem. Measure the open space.

Teacher: Assign teams one zone to survey and demonstrate a labeled top-view redesign sketch.

16-24 min | Draw the redesign

Students: Name the user. Draw the improvement on grid paper. Label the main measurements.

Teacher: Run a short gallery pitch so teams can test whether their need, location, and idea are clear.

24-28 min | Model one part

Students: Build one key feature from paper. Check its size against the drawing.

28-31 min | Pack the kit

Students: Return each measuring tool. Place the model in the class tray.

31-33 min | Return inside

Students: Walk back with your design team.

33-35 min | Exit check

Students: Write one reason the redesign helps.

What to collect or hear

Your redesign with the user need and the location marked.

Success check

Show the current-space evidence, identify the user need, and explain how the redesign improves that exact location.

Support and stretch

Grade 5: Name one user need and sketch one change.

Grade 6: Use measurements to show the change fits the space.

Another way to do it

Use photos or a map of the space indoors.

Optional next challenge

One Better Detail: Add one sign, storage, seating, shade, or habitat detail and make a short materials list for it.

Cleanup

Return measuring tools and clipboards, sort model scraps, and leave the surveyed outdoor zone unchanged.

Teacher notes

Good redesign begins with what the space already does well and what gets in the way.

A useful idea names the user, the need, the location, and how it could actually work.

Revisit the outdoor boundaries and model observing a space without disturbing it.

Assign teams one zone to survey and demonstrate a labeled top-view redesign sketch.

Run a short gallery pitch so teams can test whether their need, location, and idea are clear.

L73 | Mirror Maze

GOAL: Arrange reflective surfaces to direct a light path through checkpoints to a target.

35 minutes. Lesson reference: catalog:29

Before students arrive

Plastic mirrors and low-power lights

Target cards

Materials

Plastic mirrors; Low-power classroom lights; Target cards; Grid paper; Mirror stands; Pencils

Safety

Never shine any light toward eyes, reflective clothing, windows, or outside the marked table zone.

Run the period

00-03 min | Get ready

Students: Place the light at the fixed source mark. Keep the power off.

Teacher: Light changes direction when it reflects, and the surface angle controls the new path. A good setup reaches the target without aiming light toward people.

03-07 min | Light safety

Students: Keep the light inside the table zone. Point no light toward a face. Point no reflection toward a window. Turn off the light before moving it.

Teacher: Model one safe reflection using a low-power classroom light.

07-11 min | Predict the path

Students: Mark the source. Mark each checkpoint. Draw the expected light path.

Teacher: Show how to trace the incoming and outgoing path on paper.

11-19 min | Place the mirrors

Students: Set one plastic mirror in its stand. Match its angle to the drawing. Repeat for each mirror.

Teacher: Approve the fixed light position before teams begin.

19-26 min | Trace the light

Students: Turn on the fixed light. Trace the actual path on the grid. Mark the first missed checkpoint.

26-30 min | Adjust the angles

Students: Turn off the light. Adjust one mirror. Retest the full path.

30-32 min | Power down

Students: Turn off the light. Return mirrors to the padded tray.

32-35 min | Exit check

Students: Draw the final path. Circle the changed mirror.

What to collect or hear

The light path traced to the target and one angle fix.

Success check

Trace the successful light path and explain one mirror-angle revision.

Support and stretch

Grade 5: Hit the target with two mirrors.

Grade 6: Use one less mirror and explain the angles.

Another way to do it

Trace paths with a ruler on paper using angle cards.

Optional next challenge

One Less Mirror: Reach the same target with fewer reflective surfaces.

Cleanup

Switch off lights, handle mirrors by the edges, and return every mirror to a padded slot.

Teacher notes

Light changes direction when it reflects, and the surface angle controls the new path.

A good setup reaches the target without aiming light toward people.

Model one safe reflection using a low-power classroom light.

Show how to trace the incoming and outgoing path on paper.

Approve the fixed light position before teams begin.

L74 | Demo Day

GOAL: Independently host, demonstrate, reset, and reflect on a polished project experience for an audience.

35 minutes. Lesson reference: catalog:36

Before students arrive

Showcase exhibits or kept games

Role cards and visitor prompts

Materials

Showcase exhibits; Role cards; Safety checklist; Visitor prompts; Feedback forms; Reset supplies

Safety

Operate only approved demonstrations, keep exits and walkways clear, and pause any station that cannot reset safely.

Run the period

00-03 min | Get ready

Students: Report to your assigned station. Read your role card.

Teacher: Today the studio belongs to the teams who built it. Professional means safe, welcoming, honest about evidence, and ready to reset.

03-06 min | Station safety

Students: Operate only approved demonstrations. Keep every walkway clear. Pause any station that cannot reset safely.

Teacher: Confirm safety, traffic flow, and team roles before opening.

06-10 min | Check the station

Students: Inspect the prototype. Check the visitor boundary. Practice the emergency pause.

Teacher: Observe and record evidence without taking over student explanations.

10-19 min | Host the demo

Students: Welcome the visitor. State the design need. Demonstrate the approved action. Show the evidence.

Teacher: Lead the final reset and short reflection after the audience leaves.

19-26 min | Rotate the roles

Students: Reset after each visitor. Move to the next role card. Answer questions with evidence. Name a limit when needed.

26-30 min | Close the station

Students: Thank the final visitor. Complete the last safe reset.

30-32 min | Restore the room

Students: Return all role cards. Place every exhibit in storage position.

32-35 min | Exit check

Students: Name one strong visitor moment. Write one improvement for next time.

What to collect or hear

Your final reflection with one visitor observation.

Success check

Submit the final reflection with one audience observation, one success, and one next improvement.

Support and stretch

Grade 5: Host with a partner using role cards.

Grade 6: Host alone and change one thing after feedback.

Another way to do it

Share with a written card or by pointing to the exhibit.

Optional next challenge

Next-Year Note: Record one specific lesson that would help a future studio team start stronger.

Cleanup

Complete the full station checklist, inventory all shared materials, and leave the studio ready for its next use.

Teacher notes

Today the studio belongs to the teams who built it.

Professional means safe, welcoming, honest about evidence, and ready to reset.

Confirm safety, traffic flow, and team roles before opening.

Observe and record evidence without taking over student explanations.

Lead the final reset and short reflection after the audience leaves.